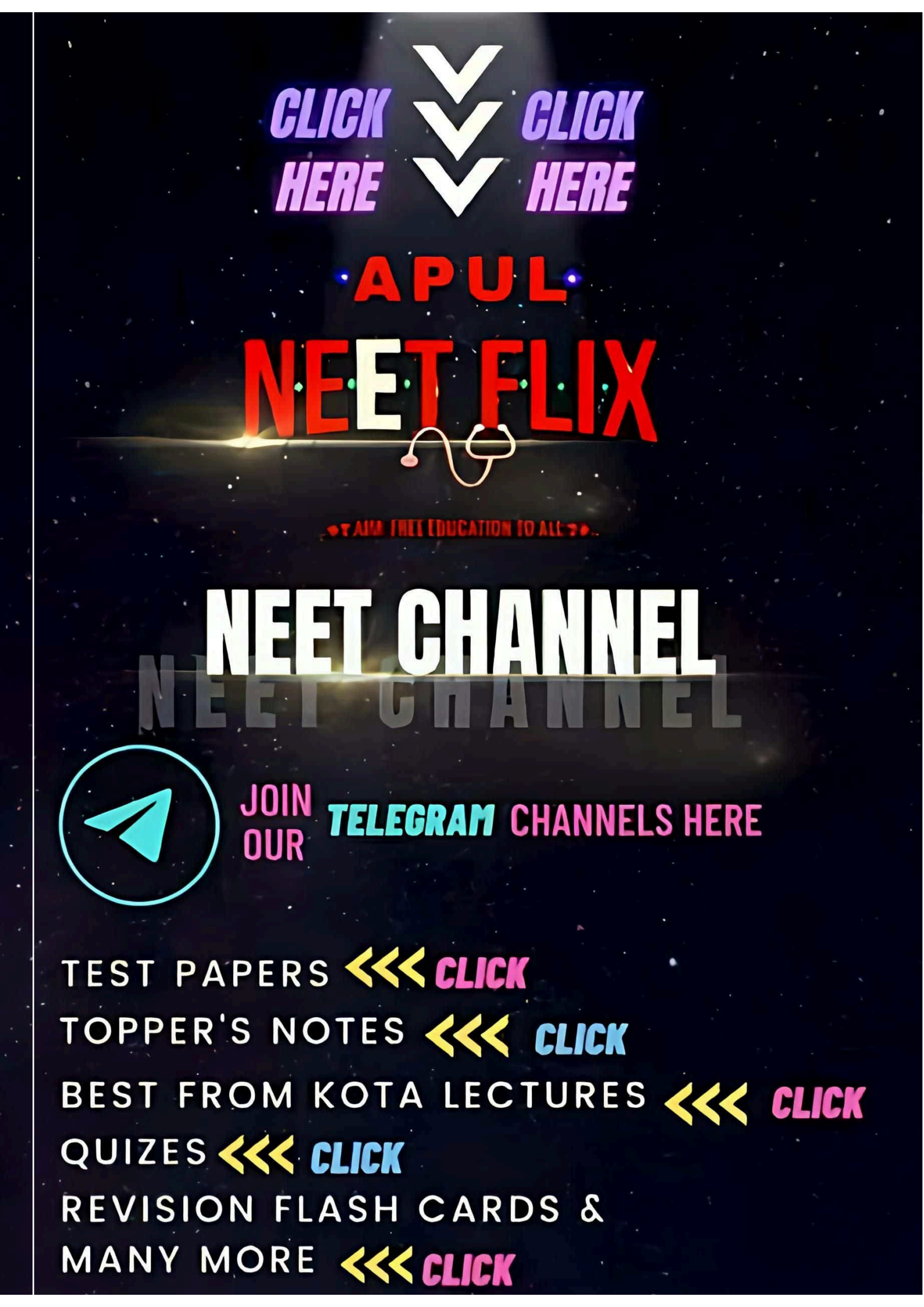

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Exercise - I Electrostatics

1. An electron is moving round the nucleus of a hydrogen atom in a circular orbit of radius r. The Coulomb force $\vec{F}$ on the electron is :-

(1) $K\frac{e^2}{r^2}\hat{r}$ (2) $-K\frac{e^2}{r^3}\hat{r}$

(3) $K\frac{e^2}{r^3}\hat{r}$ (4) $-K\frac{e^2}{r^3}\hat{r}$

PEL305

2. A charge q is placed in the middle of two equal and like point charges Q. For this system to remain in equilibrium the value of q is :–

(1) $-\frac{Q}{3}$ (2) $-\frac{Q}{4}$ (3) $\frac{Q}{2}$ (4) $-\frac{Q}{2}$

PEL306

3. Two positive charges each of equal magnitude q are placed at a separation 2a perpendicular to X-axis. Another negative charge of mass m, is placed midway between the two charges on X–axis. If this charge is displaced from equilibrium state to a distance x(x << a), then the particle :–

(1) will execute simple harmonic motion about its equilibrium position

(2) will oscillate about its equilibrium position but will not execute simple harmonic motion

(3) will not return back to the equilibrium position

(4) will stop at equilibrium position

PEL307

4. Two equal negative charges each – q, are placed at points (0, a) and (0, –a) on Y- axis, A positive charge q is released from point (2a, 0). This charge will be: -

(1) execute S.H.M. about the origin.

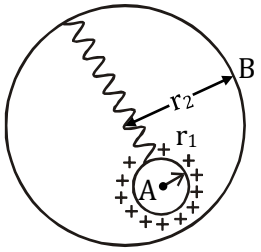
(2) oscillate but not execute S.H.M.

(3) move towards origin and will become stationary.

(4) execute S.H.M. along Y-axis.

PEL308

5. A metal sphere A of radius r_1 charged to a potential ϕ_1 is enveloped by a thin walled conducting spherical shell B of radius r_2. Then potential ϕ_2 of the sphere A after it is connected to the shell B by a thin conducting wire will be :-

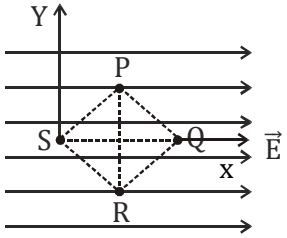


(1) $\phi_1\frac{r_1}{r_2}$ (2) $\phi_1\left(\frac{r_2}{r_1}\right)$

(3) $\phi_1\left(1-\frac{r_2}{r_1}\right)$ (4) $\phi_1\left(\frac{r_1r_2}{r_1+r_2}\right)$

PEL309

6. A point charge q moves from point P to point S along the path PQRS (figure) in a uniform electric field E pointing parallel to the positive direction of the X-axis. The co-ordinates of the points P,Q,R and S are (a, b, 0), (2a, 0, 0), (a, –b, 0) and (0, 0, 0) respectively. The work done by the field in the above process is :-



(1) qEa (2) –qEa

(3) $qEa\sqrt{2}$ (4) $qE\sqrt{[(2a)^2+b^2]}$

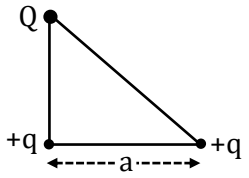
PEL310

7. Two positive point charges of 12 μC and 8 μC are 10 cm apart. The work done in bringing them 4 cm closer is :-

(1) 1.3 eV (2) 13 J (3) 5.8 J (4) 5.8 eV

PEL311

8. Three charges Q, +q and +q are placed at the vertices of a right-angled isosceles triangle as shown in figure. The net electrostatic energy of the configuration is zero. Q is equal to :-



(1) $\frac{-q}{1+\sqrt{2}}$ (2) $\frac{-2q}{2+\sqrt{2}}$ (3) –2q (4) +q

PEL312

9. A ball of mass 1g and charge 10^{-8} C moves from a point A (V_A = 600 V) to a point B whose potential is zero. Velocity of the ball at point B is 20 cms^{-1}. Velocity of the ball at point A is: –

(1) 16.7 ms^{-1} (2) 16.7 cms^{-1}

(3) 2.8 ms^{-1} (4) 2.8 cms^{-1}

PEL313

10. If a charged spherical conductor of radius 10 cm has potential V at a point distant 5 cm from its centre, then the potential at a point 15 cm away from the centre will be :-

(1) 3V (2) $\frac{3}{2}$V

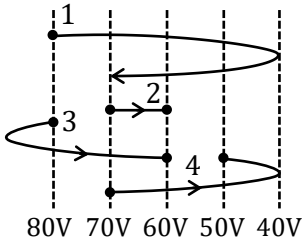
(3) $\frac{2}{3}$V (4) $\frac{1}{3}$V

PEL314

11. Figure shows a family of parallel equipotential surfaces and four paths along which an electron is made to move from one surface to another as shown.

(A) What is the direction of the electric field?

(B) Rank the paths according to work done, greatest first.



(1) Rightward; 4 > 3 > 2 > 1

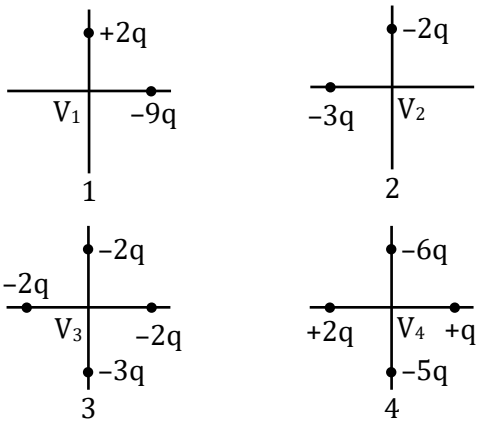
(2) Leftward; 1 > 2 > 3 > 4

(3) Rightward; 3 = 4 > 2 = 1

(4) Leftward; 1 > 2 > 3 = 4

PEL315

12. Figure given shows four arrangement of charged particles, all at the same distance from the origin. Rank the situations according to the net electric potentials (V_1, V_2, V_3, V_4) at the origin, most positive first :-



(1) $V_1 > V_2 > V_3 > V_4$ (2) $V_2 > V_1 > V_3 > V_4$

(3) $V_2 > V_1 > V_4 > V_3$ (4) $V_4 > V_1 > V_3 > V_2$

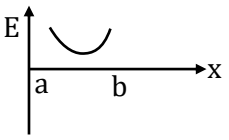
PEL316

13. There is a uniform electric field of strength 10^3 V/m along y–axis. A body of mass 1 g and charge 10^{-6} C is projected into the field from origin along the positive x–axis with a velocity 10 m/s. Its speed in m/s after 10 s is :- (neglect gravitation)

(1) 10 (2) $5\sqrt{2}$ (3) $10\sqrt{2}$ (4) 20

PEL317

14. Two point charges a & b whose magnitudes are same are positioned at a certain distance from each other; a is at origin. Graph is drawn between electric field strength E and distance x from a. E is taken positive if it is along the line joining from a to b



(1) a is positive, b is negative

(2) a & b both are positive

(3) a & b both are negative

(4) a is negative, b is positive

PEL318

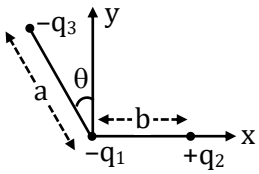
Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer	4	2	1	2	1	2	3	2	2	3	3	3	3	1

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Exercise - II Electrostatics

1. Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in the figure. The x–component of the force on $-q_1$ is proportional to: -

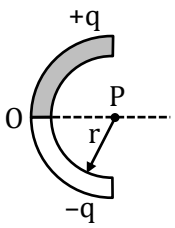


(1) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\sin\theta$ (2) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\cos\theta$

(3) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\sin\theta$ (4) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\cos\theta$

PEL319

2. A thin glass rod is bent into a semicircle of radius r. A charge +q is uniformly distributed along the upper half and a charge −q is uniformly distributed along the lower half, as shown in the figure. The magnitude and direction of the electric field $\vec{E}$ produced at P, the centre of the circle, will be :-



(1) 0

(2) $\frac{q}{\varepsilon_0\pi^2r^2}$ perpendicular to the line OP and directed downward

(3) $\frac{q}{\varepsilon_0\pi r^2}$ perpendicular to the line OP and directed downward

(4) $\frac{q}{\varepsilon_0\pi r^2}$ along the axis OP

PEL320

3. A circular wire loop of radius 'r' carries a total charge 'Q' distributed uniformly over its length. A small length dl of the wire is cut off. The electric field at the centre due to the remaining wire :-

(1) $\frac{Qd\ell}{8\pi^2\epsilon_0 r^3}$ (2) $\frac{Qd\ell}{2\pi^2\epsilon_0 r^3}$

(3) $\frac{Qd\ell}{8\pi\epsilon_0 r^3}$ (4) $\frac{Qd\ell}{4\pi^2\epsilon_0 r^3}$

PEL321

4. A solid metallic sphere has a charge + 3Q. Concentric with this sphere is a conducting spherical shell having charge −Q. The radius of the sphere is 'a' and that of the spherical shell is 'b' (b > a). What is the electric field at a distance R (a < R < b) from the centre?

(1) $\frac{4Q}{2\pi\varepsilon_0R^2}$ (2) $\frac{3Q}{4\pi\varepsilon_0R^2}$

(3) $\frac{3Q}{2\pi\varepsilon_0R^2}$ (4) $\frac{Q}{2\pi\varepsilon_0R}$

PEL322

5. Identical charges (− q) are placed at each corners of a cube of side 'b' then E.P.E. of a charge (+ q) placed at the centre of the cube will be :-

(1) $\frac{-4\sqrt{2}q^2}{\sqrt{3}\pi\epsilon_0 b}$ (2) $\frac{-8\sqrt{2}q^2}{\sqrt{3}\pi\epsilon_0 b}$

(3) $\frac{-4q^2}{\sqrt{3}\pi\epsilon_0 b}$ (4) $\frac{8\sqrt{2}q^2}{4\pi\epsilon_0 b}$

PEL323

6. A spherical conducting shell of radius R has a charge q. A point charge Q is placed at its centre. The electrostatic potential at point P at a distance $\frac{R}{2}$ from the centre of the shell is :-

(1) $\frac{(q+Q)}{4\pi\varepsilon_0}\frac{2}{R}$ (2) $\frac{2Q}{4\pi\varepsilon_0R}$

(3) $\frac{2Q}{4\pi\varepsilon_0R}-\frac{2q}{4\pi\varepsilon_0R}$ (4) $\frac{2Q}{4\pi\varepsilon_0R}+\frac{q}{4\pi\varepsilon_0R}$

PEL324

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7. A point charge q of mass m is located at the centre of a ring having radius R and charge Q with its axis oriented along X- axis. When the point charge is displaced slightly, it accelerates along the X–axis to infinity; the ultimate speed of the point charge (consider no energy loss in radiation) is :-

(1) $\sqrt{\frac{2kQq}{mR}}$ (2) $\sqrt{\frac{kQq}{mR}}$

(3) $\sqrt{\frac{kQq}{2mR}}$ (4) zero

PEL325

8. An elementary particle of mass m and charge +e is projected with velocity v towards a much more massive particle of charge Ze, where Z > 0. What is the closest possible distance of approach of the incident particle?

(1) $\frac{Ze^2}{2\pi\varepsilon_0 mv^2}$ (2) $\frac{Ze}{4\pi\varepsilon_0 mv^2}$

(3) $\frac{Ze^2}{8\pi\varepsilon_0 mv^2}$ (4) $\frac{Ze}{8\pi\varepsilon_0 mv^2}$

PEL326

9. Consider the following statements about electric dipole and select the correct ones :-

S1 : Electric dipole moment vector $\vec{p}$ is directed from negative charge to positive charge.

S2 : The electric field of a dipole at a point with position vector $\vec{r}$ depends on $|\vec{r}|$ as well as the angle between $\vec{r}$ and $\vec{p}$.

S3 : The electric dipole potential falls off as $\frac{1}{r^2}$ and not as $\frac{1}{r}$.

S4 : In a uniform electric field, the electric dipole experiences no net forces but a torque $\vec{\tau}=\vec{p}\times\vec{E}$.

(1) S2, S3 and S4

(2) S3 and S4

(3) S2 and S3

(4) all four

PEL327

Exercise-II **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9
Answer	3	2	1	2	3	4	1	1	4

Exercise - I Current Electricity

1. When a potential difference is applied across the ends of a linear metallic conductor: -

(1) group of free electrons are accelerated continuously from the lower potential end to the higher potential end of the conductor

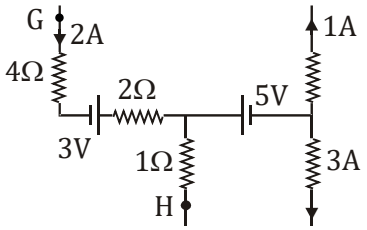
(2) group of free electrons are accelerated continuously from the higher potential end to the lower potential end of the conductor

(3) group of free electrons acquire a constant drift velocity from the lower potential end to the higher potential end of the conductor.

(4) each free electron is set in motion from their position of rest.

PCE302

2. In the part of a circuit shown in figure, the potential different ($V_G - V_H$) between points G and H will be:-

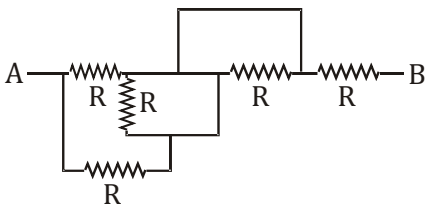


(1) 0 (2) 15 V

(3) 7 V (4) 3 V

PCE303

3. What is the equivalent resistance between A and B?

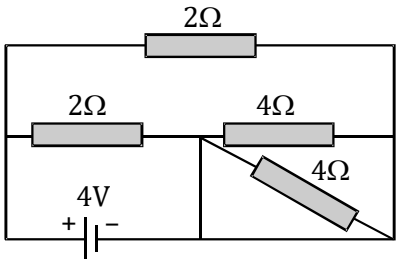


(1) $\frac{3R}{2}$ (2) 2R

(3) R (4) $\frac{R}{2}$

PCE304

4. Electric current through the battery is: -



(1) 4 A (2) 3 A

(3) 2 A (4) 0 A

PCE305

5. 100 cells, each of e.m.f. 5V and internal resistance 1Ω, are to be arranged so as to drive maximum current in a 25Ω resistance. Each row is to contain equal number of cells. The number of rows should be: -

(1) 2 (2) 4 (3) 5 (4) 10

PCE306

6. Two nonideal batteries are connected in parallel; consider the following statements :

(A) The equivalent emf is smaller than either of the two emfs.

(B) The equivalent internal resistance is smaller than either of the two internal resistances.

(1) both A and B are correct

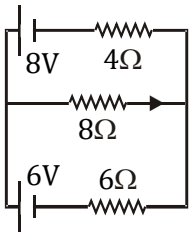
(2) A is correct but B is wrong

(3) B is correct but A is wrong

(4) both A and B are wrong

PCE307

7. The current in 8Ω resistance is (as per given circuit)



(1) 0.69 A (2) 0.92 A

(3) 1.30 A (4) 1.6 A

PCE308

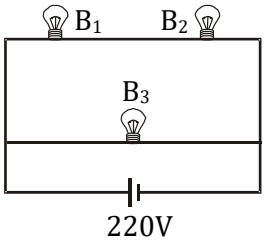
8. n identical cells, each of emf E and internal resistance r, are joined in series to form a closed circuit. One of the cell A is joined with reversed polarity. The potential difference across each cell except A, is :-

(1) $\frac{2\varepsilon}{n}$ (2) $\frac{n-1}{n}\varepsilon$

(3) $\frac{n-2}{n}\varepsilon$ (4) $\frac{2n}{n-2}\varepsilon$

PCE309

9. A 100 W bulb B_1 and two 60 W bulbs B_2 and B_3, are connected to a 220 V source, as shown in Figure. Now P_1, P_2 and P_3 are the output powers of the bulbs B_1, B_2 and B_3 respectively. Then:

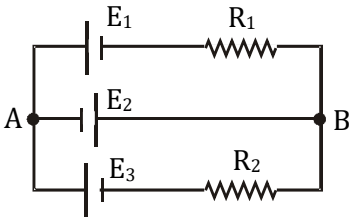


(1) $P_1 > P_2 = P_3$ (2) $P_1 > P_2 > P_3$

(3) $P_1 < P_2 = P_3$ (4) $P_1 < P_2 < P_3$

PCE310

10. In the circuit shown here, $E_1 = E_2 = E_3 = 2V$ and $R_1 = R_2 = 4$ ohms. The current flowing between points A and B through battery E_2 is:



(1) zero (2) 2 A from A to B

(3) 2 A from B to A (4) none of the above

PCE311

11. For a cell, the terminal potential difference is 2.2 V when circuit is open which reduces to 1.8 V when it is connected to a resistance of R = 5 Ω ; then internal resistance of the cell is: –

(1) $\frac{10}{9}\Omega$ (2) $\frac{9}{10}\Omega$

(3) $\frac{11}{9}\Omega$ (4) $\frac{5}{9}\Omega$

PCE312

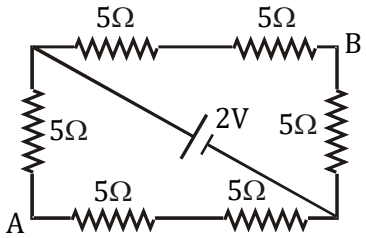
12. n identical cells whether joined together in series or in parallel, give the same current, when connected to an external resistance 'R'. The internal resistance of each cell is :–

(1) r = nR (2) r = R

(3) r = $\frac{R}{n}$ (4) r = n^2R

PCE313

13. The potential difference between the points A and B in the following circuit shown in the figure is :-



(1) $\frac{2}{3}$volts (2) $\frac{4}{5}$volts

(3) $\frac{8}{9}$volts (4) 2 volts

PCE314

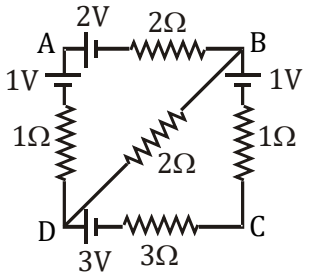
Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	
Answer	3	3	1	1	1	3	1	1	4	2	1	2	1	

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Exercise - II Current Electricity

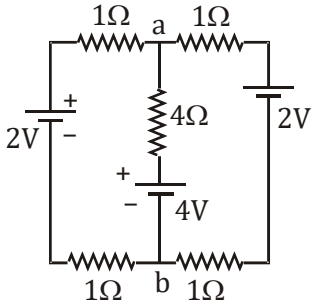
1. In the circuit shown in figure, cells of emf 2, 1, 3 and 1V, respectively having resistances 2Ω, 1Ω, 3Ω and 1Ω are their internal resistances respectively. The potential difference between D and B. (in volts)



(1) $\frac{5}{13}$ (2) $\frac{2}{13}$ (3) $\frac{10}{13}$ (4) $\frac{7}{13}$

PCE315

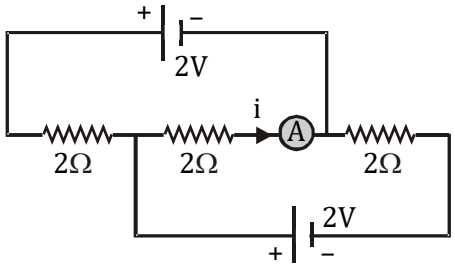
2. For the circuit shown in figure, the potential difference between points a and b is :-



(1) 2.4 V (2) 2.8 V (3) 2 V (4) 3.2 V

PCE316

3. Reading of ammeter for the following circuit is :-



(1) 1 A (2) $\frac{1}{2}$ A

(3) $\frac{2}{3}$ A (4) 3 A

PCE317

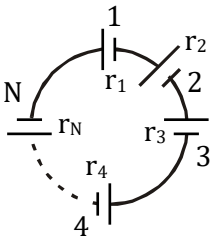
4. In Exercise-1, question no. 8 the potential difference across A is: -

(1) $\frac{2\varepsilon}{n}$ (2) $\varepsilon\left(1-\frac{1}{n}\right)$

(3) $2\varepsilon\left(1-\frac{1}{n}\right)$ (4) $\varepsilon\left(\frac{n-2}{n}\right)$

PCE318

5. A group of N cells each of whose emf varies directly with the internal resistance as per the equation $E_N = 1.5\ r_N$ are connected as shown in the figure. The current in the circuit is :-



(1) 5.1 A (2) 0.51 A
(3) 1.5 A (4) 0.15 A

PCE319

6. Two wires of resistance R_1 and R_2 at 0° C have temperature coefficients of resistance α_1 and α_2 respectively. These are joined in series. The effective temperature coefficient of resistance is: -

(1) $\frac{\alpha_1+\alpha_2}{2}$ (2) $\sqrt{\alpha_1\alpha_2}$

(3) $\frac{\alpha_1R_1+\alpha_2R_2}{R_1+R_2}$ (4) $\frac{\sqrt{R_1R_2\alpha_1\alpha_2}}{\sqrt{R_1^2+R_2^2}}$

PCE320

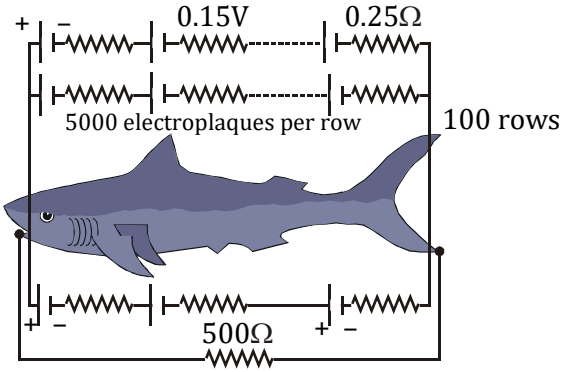
7. A galvanometer of 50 ohms resistance has 25 divisions. A current of 4×10^{-4} amperes gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of :–

(1) 2450 Ω in parallel (2) 2550 Ω in series
(3) 2450 Ω in series (4) 2500Ω in parallel

PCE321

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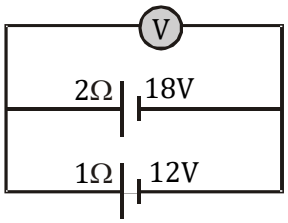
8. Eels are able to generate current with biological cells called electroplaques. The electroplaques in an eel are arranged in 100 rows, each row stretching horizontally along the body of the fish containing 5000 electroplaques. The arrangement is suggestively shown below. Each electroplaques has an emf of 0.15 V and internal resistance of 0.25 Ω. The water surrounding the eel completes a circuit between its head and its tail. If the water surrounding it has a resistance of 500 Ω, the current an eel can produce in water is about: -



(1) 1.5 A
(2) 3.0 A
(3) 15 A
(4) 30 A

PCE322

9. Two batteries, one of emf 18 volts and internal resistance 2Ω and the other of emf 12 volts and internal resistance 1Ω, are connected as shown. The voltmeter V will record a reading of : –



(1) 18 volts
(2) 30 volts
(3) 14 volts
(4) 15 volts

PCE323

Exercise-II **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9
Answer	3	1	3	3	3	3	3	1	3

Exercise - I Capacitor

1. An air capacitor, a capacitor with a dielectric and a capacitor with a conducting slab (thickness one half the separation between the plates of parallel plate air capacitor are introduced in both cases) has capacity C_1, C_2 and C_3 respectively then:

(1) $C_1 > C_2 > C_3$ (2) $C_2 > C_3 > C_1$

(3) $C_3 > C_2 > C_1$ (4) $C_3 > C_1 > C_2$

PCP183

2. A capacitance of value 4 μF charged to 50 V is connected with another capacitance of value 2μF charged to 100 V, in such a way that plates of similar charges are connected together. The total energy in multiples of 10^{-2} J before joining and after joining will be: -

(1) 1.5 and 1.33 (2) 1.33 and 1.5

(3) 3.0 and 2.67 (4) 2.67 and 3.0

PCP184

3. A 3 μF capacitor is charged to a potential of 300 V and 2 μF capacitor is charged to 200 V. The capacitors are then connected in parallel with plates of opposite polarities joined together. What amount of charge will flow, when the plates are so connected?

(1) 1300 μC (2) 800 μC

(3) 600 μC (4) 300 μC

PCP185

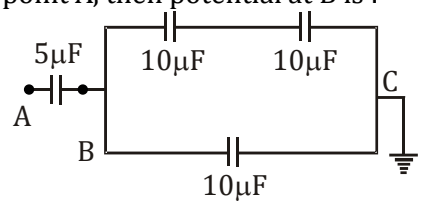
4. Two parallel plate capacitors whose capacitances are C and 2 C respectively, are joined in parallel. These are charged to V potential difference. If the battery is now removed and a dielectric of dielectric constant K is filled in between the plates of the capacitor C, then what will be the potential difference across each capacitor?

(1) $\frac{V}{K+2}$ (2) $\frac{2V}{K+2}$

(3) $\frac{3V}{K+2}$ (4) $\frac{(2+K)V}{3}$

PCP186

5. In the given circuit if point C is connected to the earth and a potential of + 2000 V is given to point A, then potential at B is :

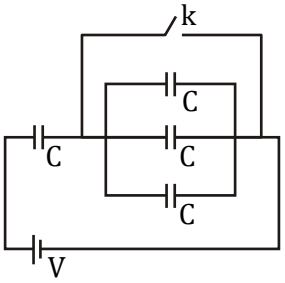


(1) 1500 V (2) 1000 V

(3) 500 V (4) 400 V

PCP187

6. The charge flowing through the cell on closing the key k is equal to:



(1) $\frac{CV}{4}$ (2) 4CV (3) $\frac{4}{3}$CV (4) $\frac{3}{4}$CV

PCP188

7. An air capacitor of capacity C = 10 μF is connected to a constant voltage battery of 12 V. Now the space between the plates is filled with a liquid of dielectric constant 5. The additional charge that flows now, from the battery to the capacitor is :

(1) 120 μC (2) 600 μC

(3) 480 μC (4) 24 μC

PCP189

8. Minimum number of 8 μF and 250 V capacitors required to make a combination of 16 μF and 1000 V are :

(1) 32 (2) 16 (3) 8 (4) 4

PCP190

9. In a capacitor of capacitance 20 μF the distance between the plates is 2 mm. If a dielectric slab of width 1 mm and dielectric constant 2 is inserted between the plates, then the new capacitance will be :

(1) 22 μF (2) 26.6 μF

(3) 52.2 μF (4) 13 μF

PCP191

10. If the distance between plates of a capacitor having capacity C & charge Q is doubled then the work done will be :

(1) $Q^2/4C$ (2) $Q^2/2C$

(3) Q^2/C (4) $2Q^2/C$

PCP192

11. A slab of copper of thickness $\frac{d}{2}$ is introduced between the plates of a parallel plate capacitor where d is the separation between its two plates. If the capacitance of the capacitor without the copper slab is C and with copper slab is C' then $\frac{C'}{C}$ is :-

(1) $\sqrt{2}$ (2) 2 (3) 1 (4) $\frac{1}{\sqrt{2}}$

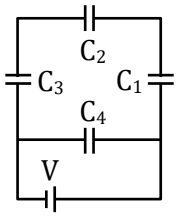
PCP193

12. Two parallel plates of the same metal and same area are placed between the plates of a parallel plate capacitor of capacity C, If the thickness of each plate is equal to $\frac{1}{5}$ th of the distance between the plates of the original capacitor then the capacity of the new capacitor is :-

(1) $\frac{5}{3}C$ (2) $\frac{3}{5}C$ (3) $\frac{3}{10}C$ (4) $\frac{10}{3}C$

PCP194

13. A network of four capacitors of capacity equal to $C_1 = C$, $C_2 = 2C$, $C_3 = 3C$ and $C_4 = 4C$ are connected to a battery as shown in the figure. The ratio of the charges on C_2 and C_4 is :–



(1) $\frac{7}{4}$ (2) $\frac{22}{3}$

(3) $\frac{3}{22}$ (4) $\frac{4}{7}$

PCP195

14. Two capacitors, 3μF and 4 μF, are individually charged across a 6 V battery. After being disconnected from the battery, they are connected together with the negative plate of one connected to the positive plate of the other. What is the final total energy stored?

(1) 1.26×10^{-4} J

(2) 2.57×10^{-4} J

(3) 1.26×10^{-6} J

(4) 2.57×10^{-6} J

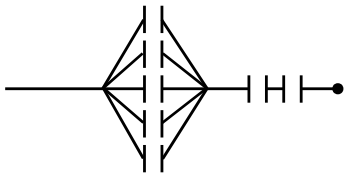
PCP196

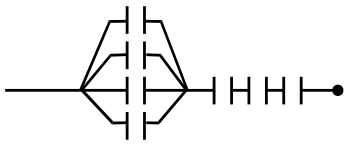
Exercise-I **ANSWER KEY**

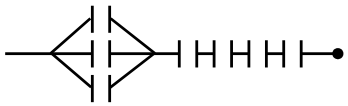
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	3	1	3	3	3	1	3	1	2	2	2	1	3	4	

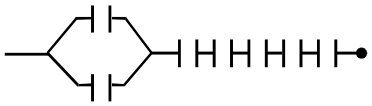
Exercise - II Capacitor

1. Seven capacitors each of capacitance 2 μF are to be connected so as to have a total capacity of $\frac{10}{11}$μF . Which will be the combination shown?

(1) 

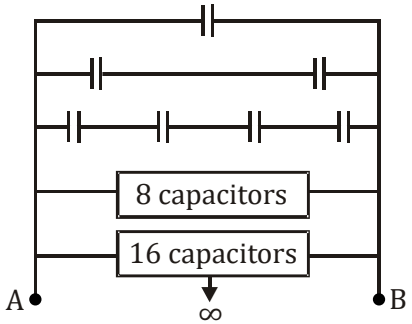
(2) 

(3) 

(4) 

PCP197

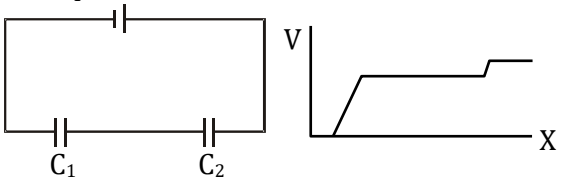
2. An infinite number of identical capacitors each of capacitance 1 μF are connected as in the adjoining figure. Then the equivalent capacitance between A and B is :



(1) 1 μF (2) 2 μF (3) $\frac{1}{2}$μF (4) ∞

PCP198

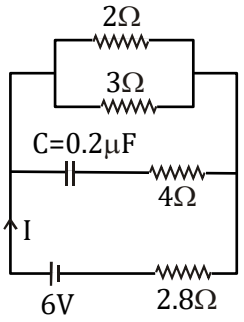
3. Figure shows two capacitors connected in series and joined to a battery. The graph shows the variation in potential as one moves from left to right along the branch containing the capacitors. Then:



(1) $C_1 > C_2$
(2) $C_1 = C_2$
(3) $C_1 < C_2$
(4) The information is not sufficient to decide the relation between C_1 and C_2

PCP199

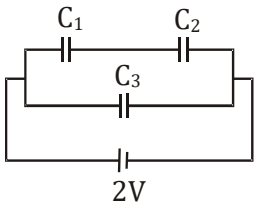
4. In the given figure the steady state current I is :



(1) zero (2) 0.6 A (3) 0.9 A (4) 1.5 A

PCP200

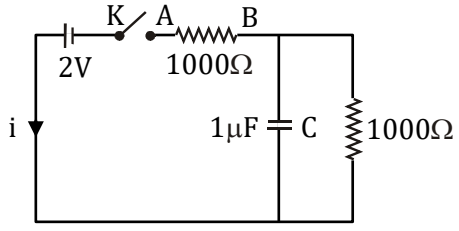
5. Two capacitors C_1 = 2 μF and C_2 = 6 μF in series, are connected in parallel to a third capacitor C_3 = 4 μF. This arrangement is then connected to a battery of e.m.f. = 2 V, as shown in the figure. How much energy is given by the battery in charging the capacitors?



(1) 22×10^{-6} J (2) 11×10^{-6} J
(3) $\left(\frac{32}{3}\right) \times 10^{-6}$ J (4) $\left(\frac{16}{3}\right) \times 10^{-6}$ J

PCP201

6. When the key is pressed at time t = 0 then which of the following statements about the current i in the resistor AB of the given circuit is true?



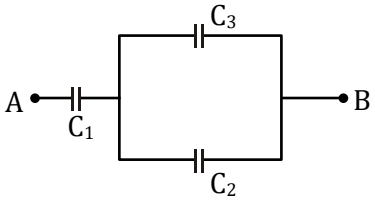
(1) at t = 0, i = 2 mA and it reduces to 1 mA at t = ∞
(2) i oscillates between 1 mA and 2 mA
(3) i = 2 mA at all time
(4) i = 1 mA at all time

PCP202

7. The combination of capacitors with C_1=3 μF, C_2 = 4 μF and C_3 = 2 μF is charged by connecting AB to a battery. Consider the following statements:

(I) Energy stored in C_1 = Energy stored in C_2+ Energy stored in C_3
(II) Charge on C_1= Charge on C_2 + Charge on C_3
(III) Potential drop across C_1 = Potential drop across C_2 = Potential drop across C_3

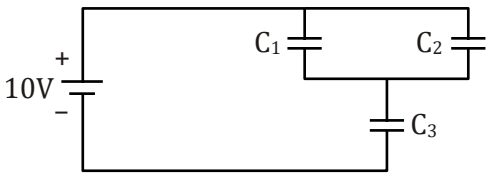
Which of these is are/correct?



(1) I and II (2) only II
(3) I and III (4) only III

PCP203

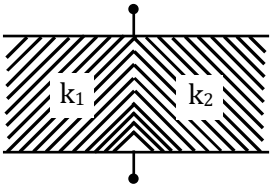
8. A 10 volts battery is connected to three capacitors; C_1 = 2 μF, C_2 = 3 μF and C_3 = 5 μF, as shown. The charges on the capacitors C_1, C_2 and C_3 are respectively: -



(1) 2 μC, 3 μC, 5 μC (2) 5 μC, 10 μC, 15 μC
(3) 10 μC, 15 μC, 25 μC (4) 4 μC, 6 μC, 10 μC

PCP204

9. Two materials of dielectric constants k_1 and k_2 are introduced to fill the space between the two parallel plates of a capacitor as shown in figure. The capacity of the capacitor is :



(1) $\frac{A\epsilon_0(k_1+k_2)}{2d}$ (2) $\frac{2A\epsilon_0}{d}\left(\frac{k_1k_2}{k_1+k_2}\right)$
(3) $\frac{A\epsilon_0}{d}\left(\frac{k_1k_2}{k_1+k_2}\right)$ (4) $\frac{A\epsilon_0}{2d}\left(\frac{k_1+k_2}{k_1k_2}\right)$

PCP205

10. An uncharged capacitor having capacitance C is connected across a battery of emf V. Now the capacitor is disconnected and then reconnected across the same battery but with reversed polarity. Then which of the statements is incorrect?

(1) After reconnecting, charge delivered by the battery is 2CV.
(2) After reconnecting, no energy is supplied by battery.
(3) After reconnecting, whole of the energy supplied by the battery is converted into heat.
(4) After reconnecting, thermal energy produced in the circuit will be equal to $2CV^2$.

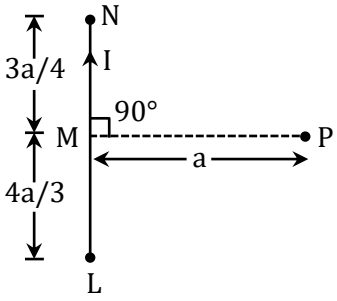
PCP206

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	2	3	4	1	1	2	3	1	2

Exercise - I Magnetic Effect of Current and Magnetism

1. Magnetic field at point 'P' due to finite length wire LN, (where I in ampere and a in meter) is :



(1) $\frac{7\mu_0 I}{10\pi a}$ (2) $\frac{14\mu_0 I}{5\pi a}$

(3) $\frac{7\mu_0 I}{20\pi a}$ (4) $\frac{70\mu_0 I}{2\pi a}$

PME358

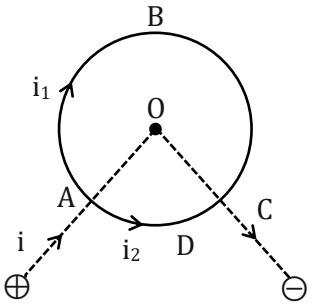
2. The effective length of a magnet is 31.4 cm and its pole strength is 0.5 Am. The magnetic moment, if it is bent in the form of a semicircle will be

(1) 0.1 Am^2 (2) 0.01 Am^2

(3) 0.2 Am^2 (4) 1.2 Am^2

PME359

3. A circular conducting loop is connected by a battery according to figure. The length of ABC and ADC is ℓ_1 and ℓ_2 respectively then which is correct : -

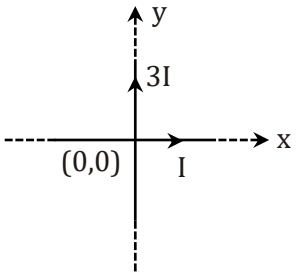


(1) $i_1/\ell_1 = i_2/\ell_2$ (2) $\frac{\ell_1^2}{i_1^2} = \frac{\ell_2^2}{i_2^2}$

(3) $i_1\ell_1 = i_2\ell_2$ (4) $i_1 i_2 = \ell_1\ell_2$

PME360

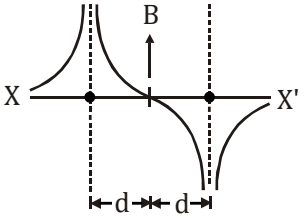
4. The equation of line on which magnetic field is zero due to system of two perpendicular infinitely long current carrying straight wires, is : -

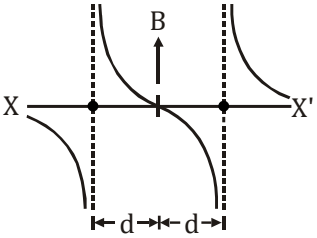


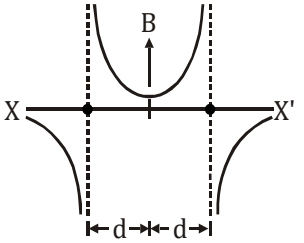
(1) x = y (2) x = 2y (3) x = 3y (4) 3x = y

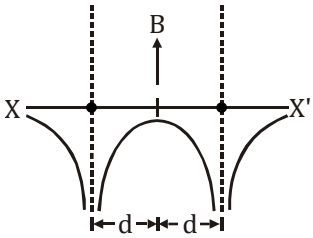
PME361

5. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper, as shown in figure. The variation of the magnetic field B along the line XX' is given by :-

(1) 

(2) 

(3) 

(4) 

PME362

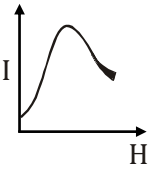
6. A solenoid of length 0.4 m, having 500 turns and 3A current flows through it. A coil of radius 0.01 m and have 10 turns and carries current of 0.4 A has to placed such that its axis is perpendicular to the axis of solenoid, then torque on coil will be:-

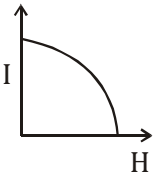
(1) 5.92×10^{-6} N.m (2) 5.92×10^{-5} N.m

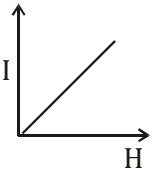
(3) 5.92×10^{-4} N.m (4) 0.592×10^{-3} N.m

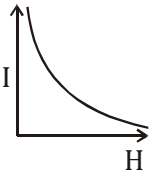
PME363

7. The correct I–H curve for paramagnetic materials is :-

(1) 

(2) 

(3) 

(4) 

PME364

8. The magnetic susceptibility of a paramagnetic substance is 3×10^{-4}. It is placed in a magnetising field of 4×10^4 A/m. The intensity of magnetisation will be:-

(1) 3×10^8 A/m (2) 12×10^8 A/m

(3) 12 A/m (4) 24 A/m

PME365

9. The magnetic susceptibility of a magnetic material is 30×10^{-4}. Its relative permeability will be: -

(1) 31×10^{-4} (2) 1.003

(3) 1.0003 (4) 29×10^{-4}

PME366

10. The magnetic moment of a magnet of mass 75 gm is 9×10^{-7} A-m^2. If the density of the material of magnet is 7.5×10^3 kg/m^3 then intensity of magnetisation will be :-

(1) 0.9 A/m (2) 0.09 A/m

(3) 9 A/m (4) 90 A/m

PME367

11. The area of hysteresis loop of a material is equivalent to 250 Joule/m^3. When 10 kg material is magnetised by an alternating field of 50Hz then energy lost in one hour will be if the density of material is 7.5 gm/cm^3.

(1) 6×10^4 Joule (2) 6×10^4 Erg

(3) 3×10^2 Joule (4) 3×10^2 Erg

PME368

12. A closely wound solenoid 80 cm long has 5 layers of winding of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, the magnitude of $\vec{B}$ inside the solenoid near its centre will be

(1) $8\pi \times 10^{-3}$ T (2) $6\pi \times 10^{-3}$ T

(3) $4\pi \times 10^{-3}$ T (4) $3\pi \times 10^{-3}$ T

PME369

13. A circular coil placed in a uniform magnetic field. If alternating current flows through this coil then :-

(1) net force will act on coil

(2) coil will be stationary

(3) coil will be rotated

(4) coil will move on straight line

PME370

14. A superconductor exhibits perfect :-

(1) Ferrimagnetism

(2) Antierromagnetism

(3) Paramagnetism

(4) Diamagnetism

PME371

15. Which of the following statements is incorrect about hysteresis :-

(1) This effect is common to all ferromagnetic substances

(2) The hysteresis loop area is proportional to the thermal energy developed per unit volume of the material

(3) The hysteresis loop area is independent of the thermal energy developed per unit volume of the material

(4) The shape of the hysteresis loop is characteristic of the material

PME372

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	3	3	2	1	3	3	2	2	1	1	2	4	3

AIM: FREE EDUCATION TO ALL APUL

Exercise - II Magnetic Effect of Current and Magnetism

1. A long horizontal wire 'A' is rigidly fixed and an another wire 'B' which is placed directly below and parallel to wire 'A'. Wire 'B' remains suspended in air due to magnetic attraction. If direction of current is reversed in any one wire then due to gravity instantaneous acceleration of free wire 'B' (where g is acceleration due to gravity)

(1) g, in downward direction

(2) g, in upward direction

(3) 2g, in downward direction

(4) 2g, in upward direction

PME373

2. H^+, He^+ and O^{++} all having the same kinetic energy pass through a region with is a uniform magnetic field perpendicular to their velocity. The masses of H^+, He^+ and O^{++} are 1 amu, 4 amu and 16 amu respectively then :-

(a) H^+ will be deflected most

(b) O^{++} will be deflected most

(c) He^+ and O^{++} will be deflected equally

(d) all will be deflected equally

(1) a, b (2) a, b, c

(3) only a (4) a, c

PME374

3. A coil in the shape of equilateral triangle of side 0.02 m is suspended from the vertex such that it is hanging in a vertical plane between the pole-pieces of a permanent magnet producing a horizontal magnetic field of 5×10^{-2} T. When a current of 0.1 A passes through it and the magnetic field is parallel to its plane then couple acting on the coil is :-

(1) 8.65×10^{-7} N-m

(2) 6.65×10^{-7} N-m

(3) 3.35×10^{-7} N-m

(4) 3.91×10^{-7} N-m

PME375

4. In a mass spectrograph O^{++}, C^+, He^{++} and H^+ are projected on a photographic plate with same velocity in uniform magnetic field then which will strike the plate farthest :-

(1) O^{++} (2) C^+ (3) He^{++} (4) H^+

PME376

5. The magnetic field in a certain region of space is given by $\vec{B} = 8.35 \times 10^{-2}\,\hat{i}$ T. A proton is shot into the field with velocity $\vec{v} = (2 \times 10^5\,\hat{i} + 4 \times 10^5\,\hat{j})$ m/s. The proton follows a helical path in the field. The distance moved by proton in the x-direction during the period of one revolution in the yz-plane will be

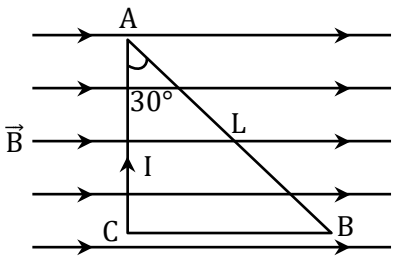
(1) 0.053 m (2) 0.136 m

(3) 0.157 m (4) 0.236 m

(Mass of proton = 1.67×10^{-27} kg)

PME377

6. Consider the inferences given below in respect of the following current loop of wire kept in a magnetic field $\vec{B}$.



A. The magnitude and direction of force on the element AC of the wire is $\frac{\sqrt{3}}{2}$ILB directed into the page.

B. The magnitude and direction of force on the element AB of the wire is $\frac{\sqrt{3}}{2}$ILB directed into the page.

C. The total force on the loop ABCA is zero.

Which of the above is /are not true?

(1) A and B (2) B only

(3) A and C only (4) B and C

PME378

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7. A planar coil having 12 turns carries 15 A current. The coil is oriented with respect to the uniform magnetic field $\vec{B}=0.2\hat{i}\,T$ such that its directed area is $\vec{A}=-0.04\hat{i}m^2$. The potential energy of the coil in the given orientation is

(1) 0

(2) + 0. 72

(3) +1.44 J

(4) –1.44 J

PME379

8. A frog can be levitated in a magnetic field produced by a current in a vertical solenoid placed below the frog. This is possible because the body of the frog behaves as: -

(1) paramagnetic.

(2) diamagnetic

(3) ferromagnetic

(4) antiferromagnetic

PME380

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	3	4	1	2	3	2	3	2

Exercise - I Electromagnetic Induction

1. A coil of copper wire is connected in series with a bulb, a battery and a switch. When the circuit is completed the bulb lights up immediately. The circuit is switched off and a rod of soft iron is placed inside the coil. On completing the circuit again. It is observed that : -

(1) Bulb is not so bright

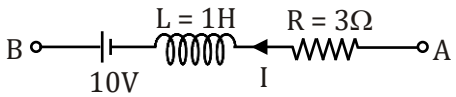
(2) There is a slight delay before bulb lights to its normal brightness

(3) The bulb is initially bright but gradually becomes dim

(4) The bulb is brighter than before

PEI221

2. In the branch AB of a circuit, as shown in the figure, a current I = (t + 2) A is flowing, where t is the time in second. At t = 0, the value of $(V_A - V_B)$ will be :-



(1) 3V (2) –3V (3) –5V (4) 5V

PEI222

3. An e.m.f. of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents at time $t \to \infty$ and at t = 1 second is: -

(1) $\frac{e^{1/2}}{e^{1/2}-1}$ (2) $\frac{e^2}{e^2-1}$

(3) $1-e^{-1}$ (4) e^{-1}

PEI223

4. If 2.2 kW power transmits 22000 volts in a line of 10Ω resistance, the value of power loss will be: -

(1) 0.1 watt (2) 14 watts

(3) 100 watts (4) 1000 watts

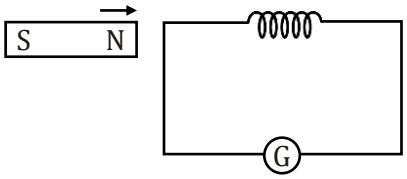
PEI224

5. A 50 mH coil carries a current of 2 ampere. The energy stored in joules is

(1) 1 (2) 0.1 (3) 0.05 (4) 0.5

PEI225

6. As shown is the figure, a magnet is brought towards a fixed coil. Due to this the induced emf, current and the charge are E, I and Q respectively. If the speed of the magnet is double then the following statement is wrong: –



(1) E increases (2) I increases

(3) Q does not change (4) Q increases

PEI226

7. The equivalent inductance of two inductances is 2.4 henry when connected in parallel and 10 henry when connected in series. The difference between the two inductances is :-

(1) 2 henry (2) 3 henry

(3) 4 henry (4) 5 henry

PEI227

8. A transformer is employed to reduce 220 V to 11 V. The primary draws a current of 5 A and the secondary 90 A. The efficiency of the transformer is

(1) 20% (2) 40% (3) 70% (4) 90%

PEI228

9. A magnet is dropped down an infinitely long vertical copper tube. Then

(1) The magnet moves with continuously decreasing velocity and ultimately comes to rest

(2) The magnet moves with continuously increasing velocity and ultimately acquires a constant terminal velocity

(3) The magnet moves with continuously increasing velocity and acceleration.

(4) The magnet moves with continuously increasing velocity but constant acceleration

PEI229

10. If a copper ring is moved quickly towards south pole of a powerful stationary bar magnet, then: -

(1) Current flows through the copper ring

(2) Voltage in the magnet increases

(3) Current flows in the magnet

(4) Copper ring will get magnetised

PEI230

11. Which of the following units denotes the dimensions ML^2/Q^2, where Q denotes the electric charge: -

(1) Weber (2) Wb/m^2

(3) henry (4) H/m^2

PEI231

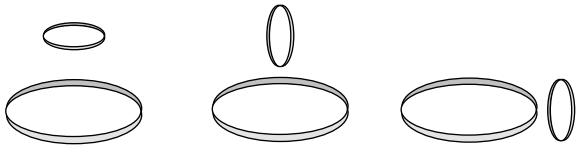
12. A small cube of volume 1 mm^3 placed at the centre of circular loop of radius of 100 cm, carrying a current 1A. The magnetic energy stored in side cube is :-

(1) $\frac{\pi}{2}\times10^{16}$ J (2) $2\pi\times10^{-7}$ J

(3) $\frac{\pi}{2}\times10^{-16}$ J (4) $\frac{\pi}{2}\times10^{-7}$ J

PEI232

13. Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be



(A) (B) (C)

(1) Maximum in situation (A)

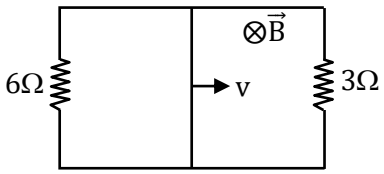
(2) Maximum in situation (B)

(3) Maximum in situation (C)

(4) The same in all situations

PEI233

14. A rectangular loop with a sliding connector of length l = 1.0 m is situated in a uniform magnetic field B = 2T perpendicular to the plane of loop. Resistance of connector is r = 2Ω. Two resistance of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity v = 2m/s is



(1) 6N (2) 4N

(3) 2N (4) 1N

PEI234

15. In cylindrical region of radius 2m magnetic field is changing as B = (2t + 5) T. Find induced electric field in N/C at r = $\frac{R}{4}$.

(1) $\frac{1}{8}$ (2) $\frac{1}{4}$

(3) $\frac{1}{2}$ (4) 0

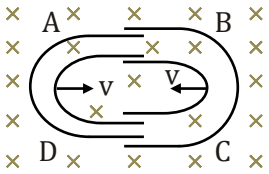
PEI235

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	1	2	4	1	4	2	1	3	3	1	3	3

Exercise - II Electromagnetic Induction

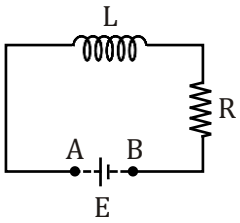
1. One conducting U tube can slide inside another as shown in figure, maintaining electrical contacts between the tubes. The magnetic field B is perpendicular to the plane of the figure. If each tube moves towards the other at a constant speed v, then the induced emf in the circuit, where ℓ is the width of each tube: -



(1) 2 Bℓv (2) Zero
(3) – Bℓv (4) Bℓv

PEI236

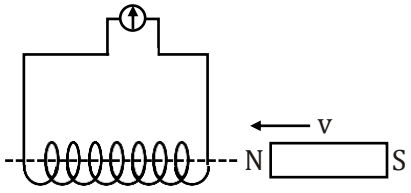
2. An inductor (L = 100 mH), a resistor (R = 100Ω) and a battery (E = 100 V) are initially connected in series as shown in the figure. After a long time the battery is disconnected after short circuiting the points A and B. The current in the circuit 1 ms after the short circuit is :-

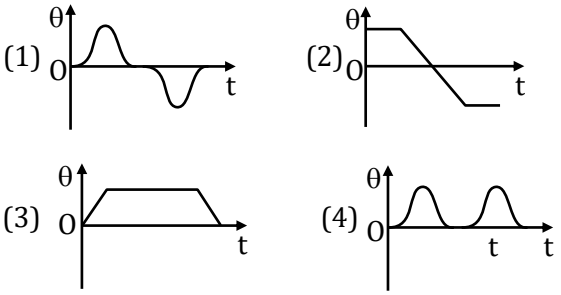


(1) 1A (2) 1/e A (3) e A (4) 0.1 A

PEI237

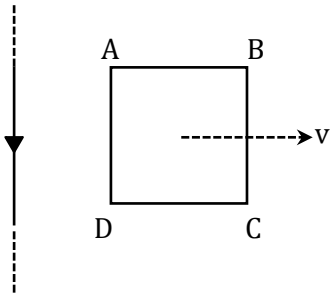
3. A short bar magnet passes at a steady speed right through a long solenoid. A galvanometer is connected across the solenoid. Which graph best represents the variation of the galvanometer deflection θ with time: -





PEI238

4. A metallic frame moves with constant velocity v near a infinite length current carrying wire, At any instant induced emf in sides AD and BC are 5V and 2V respectively. If Resistance of frame is 6Ω the magnitude and direction of induced current in it :-



(1) 2A, ACW
(2) 0.5A, ACW
(3) 2A, CW
(4) 0.5A, CW

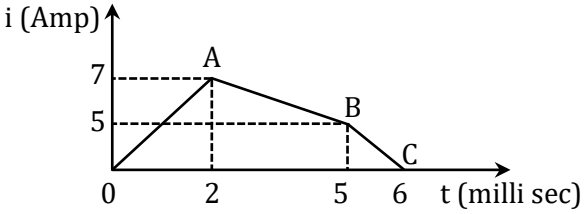
PEI239

5. The length of thin wire required to manufacture a solenoid of inductance L and length ℓ, (if the cross-sectional diameter is considered less than its length) is :-

(1) $\sqrt{\frac{\pi L\ell}{2\mu_0}}$ (2) $\sqrt{\frac{4\pi L\ell}{\mu_0}}$
(3) $\sqrt{\frac{2\pi L\ell}{\mu_0}}$ (4) $\sqrt{\frac{\pi L\ell}{\mu_0}}$

PEI240

6. The current through a 4.6 H inductor is shown in the following graph. The induced emf during the time interval t = 5 milli-sec to 6 milli-sec will be

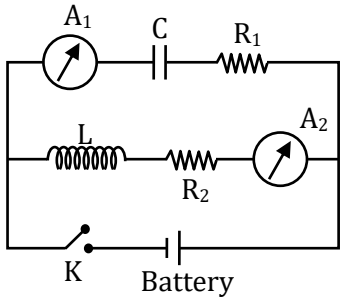


(1) 10^3 V (2) -23×10^3 V

(3) 23×10^3 V (4) Zero

PEI241

7. In a circuit inductance L and capacitance C are connected as shown in figure and A_1 and A_2 are ammeters. When key k is pressed to complete the circuit, then just after closing key k, the readings of A_1 and A_2 will be :



(1) Zero in both A_1 and A_2

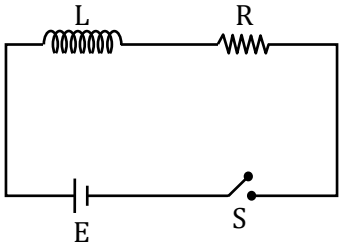
(2) Maximum in both A_1 and A_2

(3) Zero in A_1 and maximum in A_2

(4) Maximum in A_1 and zero in A_2

PEI242

8. In the circuit shown, switch S is closed at time t = 0. The charge which passes through the battery in one time constant is: -

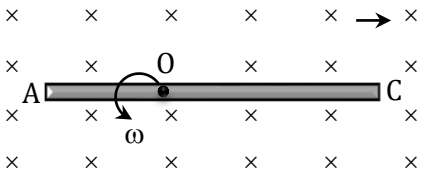


(1) $\frac{eR^2E}{L}$ (2) $E\left(\frac{L}{R}\right)$

(3) $\frac{EL}{eR^2}$ (4) $\frac{eL}{ER}$

PEI243

9. A conducting rod AC of length 4l is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. AO = l and OC = 3l. Then



(1) $V_A - V_O = \frac{B\omega l^2}{2}$

(2) $V_O - V_C = \frac{7}{2}B\omega l^2$

(3) $V_A - V_C = 4B\omega l^2$

(4) $V_C - V_O = \frac{9}{2}B\omega l^2$

PEI244

10. The number of turns of primary and secondary coils of a transformer are 5 and 10 respectively and the mutual inductance of the transformer is 25 henry. Now the number of turns in the primary and secondary of the transformer are made 10 and 5 respectively. The mutual inductance of the transformer in henry will be

(1) 6.25

(2) 12.5

(3) 25

(4) 50

PEI245

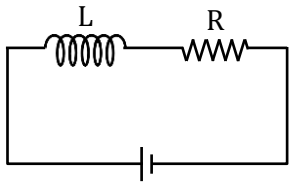
Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	2	1	2	2	3	4	3	3	3

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Exercise - I Alternating Current

1. Time constant of the given circuit is τ. If the battery is replaced by an ac source having voltage $V = V_0 \cos \omega t$, power factor of the circuit will be :-



(1) $\omega\tau$ (2) $\frac{1}{\sqrt{1+(\omega\tau)^2}}$

(3) $\sqrt{1+(\omega\tau)^2}$ (4) None

PAC190

2. An inductance L, a capacitance C and resistance R may be connected to an AC source of angular frequency ω, in three different combinations of RC, RL and RLC in series. Assume that $\omega L = \frac{1}{\omega C}$. The power drawn by the three combinations are P_1, P_2, P_3 respectively. Then :-

(1) $P_1 > P_2 > P_3$ (2) $P_1 = P_2 < P_3$

(3) $P_1 = P_2 > P_3$ (4) $P_1 = P_2 = P_3$

PAC191

3. The r.m.s. value of current for a variable current $i = i_1 \cos \omega t + i_2 \sin \omega t$:-

(1) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (2) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$

(3) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (4) $\frac{1}{2}\left(i_1^2 + i_2^2\right)^{1/2}$

PAC192

4. In an ac circuit I = 100 sin 200πt. The time required for the current to achieve its peak value will be –

(1) $\frac{1}{100}$sec (2) $\frac{1}{200}$sec

(3) $\frac{1}{300}$sec (4) $\frac{1}{400}$sec

PAC193

5. The inductance of the oscillatory circuit of a radio station is 10 milli henry and its capacitance is 0.25μF. Taking the effect of the resistance negligible, wavelength of the broadcasted waves will be (velocity of light = 3.0×10^8 m/s, $\pi = 3.14$):

(1) 9.42×10^4 m (2) 18.8×10^4 m

(3) 4.5×10^4 m (4) none of these

PAC194

6. A coil has an inductance of 0.7 henry and is joined in series with a resistance of 220 Ω. When the alternating emf of 220 V at 50 Hz is applied to it then the phase through which current lags behind the applied emf and the wattless component of current in the circuit will be respectively

(1) 30°, 1 A (2) 45°, 0.5 A

(3) 60°, 1.5 A (4) none of these

PAC195

7. In an ac circuit, the instantaneous values of e.m.f. and current are e = 200 sin 314 t volt and $i = \sin\left(314t + \frac{\pi}{3}\right)$ ampere. The average power consumed in watt is

(1) 200 (2) 100 (3) 50 (4) 25

PAC196

8. There is a 5Ω resistance in an A.C., circuit. Inductance of 0.1 H is connected with it in series. If equation of A.C. e.m.f. is 5 sin 50t then the phase difference between current and e.m.f. is :-

(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{4}$ (4) 0

PAC197

9. If an alternating current $i = i_m \sin \omega t$ is flowing through a capacitor then voltage drop ΔV_C across capacitor C will be ?

(1) $-\frac{i_m}{\omega C}\sin \omega t$ (2) $-\frac{i_m}{\omega C}\cos \omega t$

(3) $-\frac{i_m}{\omega C}\left(\sin \omega t + \frac{\pi}{4}\right)$ (4) $\frac{i_m}{\omega C}\left(\sin \omega t - \frac{\pi}{4}\right)$

PAC198

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10. If an alternating current $i = i_m \sin \omega t$ is flowing through an inductor then voltage drop ΔV_L across inductor L will be :-

(1) $i_m \omega L \sin \omega t$ (2) $i_m \omega L \cos \omega t$

(3) $i_m \omega L \sin\left(\omega t + \frac{\pi}{4}\right)$ (4) $i_m \omega L \cos\left(\omega t - \frac{\pi}{4}\right)$

PAC199

11. If frequency of alternating source is made zero then which of the following statement is true :

(1) Current through capacitor will be zero

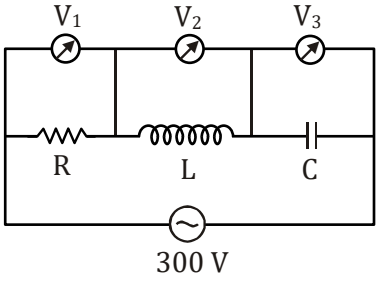
(2) Current through resistance will be zero

(3) Current through inductance will be zero

(4) All

PAC200

12. The figure shows a LCR network connected to 300 V a.c. supply. The circuit elements are such that $R = X_L = X_C = 10\Omega$. V_1, V_2 and V_3 are three a.c. voltmeters connected as shown in the figure. Which of the following represents the correct set of readings of the voltmeters?



(1) $V_1 = 100$ V, $V_2 = 100$ V, $V_3 = 100$ V

(2) $V_1 = 150$ V, $V_2 = 0$ V, $V_3 = 150$ V

(3) $V_1 = 300$ V, $V_2 = 100$ V, $V_3 = 100$ V

(4) $V_1 = 300$ V, $V_2 = 300$ V, $V_3 = 300$ V

PAC201

13. An RLC circuit has f_1 and f_2 as the half power frequency and f_0 as the resonant frequency. The Q factor of the circuit is given by

(1) $\frac{f_1 + f_2}{2f_0}$ (2) $\frac{f_1 - f_0}{f_2 - f_0}$

(3) $\frac{f_0}{f_1 - f_2}$ (4) $\frac{f_1 - f_2}{f_0}$

PAC202

14. The relation between an A.C. voltage source and time in SI units is : $V = 120 \sin(100\pi t) \cos(100\pi t)$ volt value of peak voltage and frequency will be respectively :–

(1) 120 volt and 100 Hz

(2) volt and 100 Hz

(3) 60 volt and 200 Hz

(4) 60 volt and 100 Hz

PAC203

15. Two alternating voltage generators produce e.m.f. of the same amplitude E_0 but with a phase difference of $\pi/3$. The resultant e.m.f. is-

(1) $E_0 \sin(\omega t + \pi/3)$

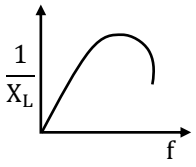
(2) $E_0 \sin(\omega t + \pi/6)$

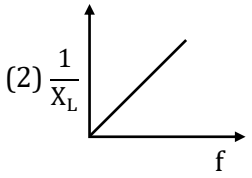
(3) $\sqrt{3}E_0 \sin(\omega t + \pi/6)$

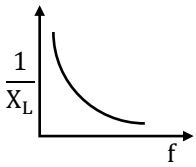
(4) $\sqrt{3}E_0 \sin(\omega t + \pi/2)$

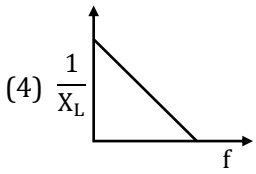
PAC204

16. In pure inductive circuit, the curves between frequency f and reciprocal of inductive reactance $1/X_L$ is

(1) 

(2) 

(3) 

(4) 

PAC205

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	4	1	2	3	3	2	2	1	4	3	4	3
Question	16														
Answer	3														

Exercise - II Alternating Current

1. The r.m.s. value of an ac of 50 Hz is 10 amp. The time taken by the alternating current in reaching from zero to maximum value and the peak value of current will be

(1) 2×10^{-2} sec and 14.14 amp

(2) 1×10^{-2} sec and 7.07 amp

(3) 5×10^{-3} sec and 7.07 amp

(4) 5×10^{-3} sec and 14.14 amp

PAC206

2. An alternating emf of angular frequency ω is applied across an inductance. The instantaneous power developed in the circuit has an angular frequency:

(1) $\omega/4$ (2) $\omega/2$

(3) ω (4) 2ω

PAC207

3. Alternating current is flowing in inductance L and resistance R. The frequency of source is $\omega/2\pi$. Which of the following statement is correct :

(1) For low frequency the limiting value of impedance is L.

(2) For high frequency the limiting value of impedance is ωL.

(3) For high frequency the limiting value of impedance is R.

(4) For low frequency the limiting value of impedance is ωL.

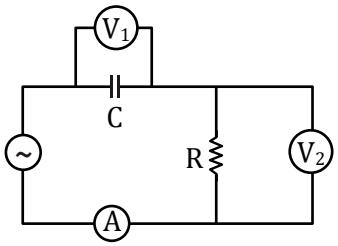
PAC208

4. An AC circuit draws 5A at 160 V and the power consumption is 600 W. Then the power factor is :-

(1) 1 (2) 0.75 (3) 0.50 (4) Zero

PAC209

5. The diagram shows a capacitor C and a resistor R connected in series to an AC source, V_1 and V_2 are voltmeters and A is an ammeter. Consider now the following statements :



(I) Readings in A and V_2 are always in phase

(II) Reading in V_1 is ahead with reading in V_2

(III) Readings in A and V_1 are always in phase

Which of these statements are is correct :

(1) I only (2) II only

(3) I and II only (4) II and III only

PAC210

6. In which of the following case power factor will be negligible :–

(1) Inductance and resistance both high

(2) Inductance and resistance both low.

(3) Low resistance and high inductance

(4) High resistance and low inductance

PAC211

7. Voltage and current in an ac circuit are given by

$V = 5\sin\left(100\pi t - \frac{\pi}{6}\right)$ and

$I = 4\sin\left(100\pi t + \frac{\pi}{6}\right)$

(1) Voltage leads the current by 30°

(2) Current leads the voltage by 30°

(3) Current leads the voltage by 60°

(4) Voltage leads the current by 60°

PAC212

8. A bulb and a capacitor are connected in series to a source of alternating current. If its frequency is increased, while keeping the voltage of the source constant, then

(1) Bulb will give more intense light.

(2) Bulb will give less intense light.

(3) Bulb will give light of same intensity as before

(4) Bulb will stop radiating light.

PAC213

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9. For series LCR resonance circuit, wrong statement is

(1) Applied e.m.f. and potential difference across resistance are in same phase

(2) Applied e.m.f. and potential difference at inductor coil have phase difference of $\pi/2$

(3) Potential difference at capacitor and inductor have phase difference of $\pi/2$

(4) Potential difference across resistance and capacitor have phase difference of $\pi/2$

PAC214

10. The current in series LCR circuit will be maximum when ω is

(1) As large as possible

(2) Equal to natural frequency of LCR system

(3) $\sqrt{LC}$

(4) $\sqrt{1/LC}$

PAC215

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	4	4	2	2	1	3	3	1	3	4

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Exercise - I — Electromagnetic Waves

1. The pressure exerted by an electromagnetic wave of intensity I W/m² on a non-reflecting surface is :-
(1) Ic (2) Ic² (3) I/c (4) I/c²
PEW070

2. A plane electromagnetic wave of frequency 40 MHz travels in free space in the X-direction. At some point and at some instant, the electric field $\vec{E}$ has its maximum value of 750 N/C in Y-direction. The wavelength of the wave is –
(1) 3.5 m (2) 5.5 m
(3) 7.5 m (4) 9.5 m
PEW071

3. In **Q.2,** the period of the wave will be –
(1) 2.5 μs (2) 0.25 μs
(3) 0.025 μs (4) none of these
PEW072

4. If c is the speed of electromagnetic waves in vacuum, its speed ν in a medium of dielectric constant K and relative permeability μ_r is (c is the speed of light)
(1) $\nu = \frac{1}{\sqrt{\mu_r K}}$ (2) $\sqrt{\mu_r c}$
(3) $\nu = \frac{c}{\sqrt{\mu_r K}}$ (4) $\nu = \frac{K}{\sqrt{\mu_r c}}$
PEW073

5. The oscillating electric and magnetic vectors of an electromagnetic wave are oriented along
(1) the same direction but differ in phase by 90°
(2) the same direction and are in phase
(3) mutually perpendicular directions and are in phase
(4) mutually perpendicular directions and differ in phase by 90°
PEW074

6. An EM wave is propagating in a medium with a velocity $\vec{v} = v\hat{i}$. The instantaneous oscillating electric field of this of EM wave is along +y axis. Then the direction of oscillating magnetic field of the EM wave will be along
(1) –x direction (2) –z direction
(3) –y direction (4) +z direction
PEW075

7. The sun delivers 10^3 W/m² of electromagnetic flux to the earth's surface. The total power that is incident on a roof of dimensions 8m × 20m, will be
(1) 6.4×10^3 W (2) 3.4×10^4 W
(3) 1.6×10^5W (4) none of these
PEW076

8. A parallel plate capacitor consists of two circular plates each of radius 2 cm, separated by a distance of 0.1 mm. Is being charged at rate of 5×10^{13} V/s, then the value of displacement current is
(1) 5.50 A (2) 5.56×10^2 A
(3) 5.56×10^3 A (4) 2.28×10^4 A
PEW077

9. A parallel plate capacitor with plate area A and separation between the plates d, is charged by a constant current i. Consider a plane surface of area A/2 parallel to the plates and drawn symmetrically between the plates, then the displacement current through this area, will be –
(1) i (2) $\frac{i}{2}$
(3) $\frac{i}{4}$ (4) None of these
PEW078

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	3	3	3	3	3	4	3	3	2

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Exercise - II — Electromagnetic Waves

1. The sun radiates electromagnetic energy at the rate of 3.9×10^{26} W. It's radius is 6.96×10^{8} m. The intensity of sun light at the solar surface will be – (in W/m^2)

(1) 1.4×10^{4}

(2) 2.8×10^{5}

(3) 4.2×10^{6}

(4) 6.4×10^{7}

PEW079

2. In. Q.1, if the distance from the sun to the earth is 1.5×10^{11} m, then the intensity of sunlight on earth's surface will be – (in W/m^2)

(1) 1.38×10^{3}

(2) 2.76×10^{4}

(3) 5.52×10^{5}

(4) none of these

PEW080

3. A long straight wire of resistance R, radius a and length l carries a constant current I. The pointing vector for the wire will be

(1) $\frac{IR}{2\pi a l}$ (2) $\frac{IR^2}{al}$

(3) $\frac{I^2R}{al}$ (4) $\frac{I^2R}{2\pi a l}$

PEW081

4. A plane electromagnetic wave of intensity of 10 W/m^2 strikes a small mirror of area 20 cm^2, held perpendicular to the approaching wave. The radiation force on the mirror will be

(1) 6.6×10^{-11} N

(2) 1.33×10^{-11} N

(3) 1.33×10^{-10} N

(4) 6.6×10^{-10} N

PEW082

Exercise-II — ANSWER KEY

Question	1	2	3	4
Answer	4	1	4	3

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Exercise - I

Ray Optics and Optical Instruments

1. A man moves towards a plane mirror with a velocity v in a direction making an angle θ with the normal to the mirror. The magnitude of velocity of the image relative to man normal to mirror will be

(1) 2 v (2) 2 v cosθ

(3) 2 v sinθ (4) 2v/cosθ

PRI279

2. A virtual image three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. The distance of the object from the mirror is -

(1) 5 cm (2) 12 cm

(3) 10 cm (4) 20 cm

PRI280

3. A convex mirror of focal length f produces an image (1/n)th of the size of the object. The distance of the object from the mirror is

(1) nf (2) f/n

(3) (n + 1)f (4) (n – 1)f

PRI281

4. A microscope is focussed on a coin lying at the bottom of a beaker. The microscope is now raised up by 1 cm. To what depth should the water be poured into the beaker so that coin is again in the focus (Refractive index of water is 4/3)

(1) 1 cm (2) $\frac{4}{3}$ cm

(3) 3 cm (4) 4 cm

PRI282

5. A vessel is half filled with a liquid of refractive index μ. The other half of the vessel is filled with an immiscible liquid of refractive index 1.5μ. The apparent depth of vessel is 50% of the actual depth. The value of μ is

(1) 1.6 (2) 1.67

(3) 1.5 (4) 1.4

PRI283

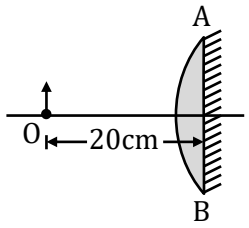
6. A clear transparent glass sphere (μ = 1.5) of radius R is immersed in a liquid of refractive index 1.25. A parallel beam of light incident on it will converge to a point. The distance of this point from the centre will be

(1) – 3R (2) + 3R

(3) – R (4) + R

PRI284

7. A point object is placed at a distance of 20 cm from a thin plano-convex lens of focal length 15 cm. If the plane surface is silvered, the image will from at



(1) 60 cm from left of AB

(2) 30 cm from left of AB

(3) 12 cm from left of AB

(4) 60 cm from right of AB

PRI285

8. A telescope consisting of objective of focal length 60cm and a single lens eye piece of focal length 5cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of 2° at the objective, then angular width of image will be.

(1) 10° (2) 24° (3) 50° (4) 1/6°

PRI286

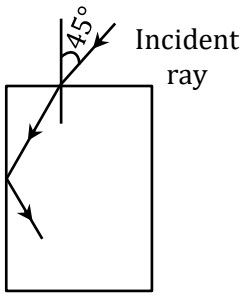
9. A film projector magnifies a film of area 100 square centimeter on screen. If linear magnification is 4 then area of magnified image on screen will be –

(1) 1600 sq.cm (2) 800 sq.cm

(3) 400 sq.cm (4) 200 sq.cm

PRI287

10. For the given incident ray as shown in figure, the condition of total internal reflection of this ray the minimum refractive index of prism will be :–



(1) $\frac{\sqrt{3}+1}{2}$ (2) $\frac{\sqrt{2}+1}{2}$

(3) $\sqrt{\frac{3}{2}}$ (4) $\sqrt{\frac{7}{6}}$

PRI288

11. A bulb is located on a wall. Its image is to be obtained on a parallel wall with the help of convex lens. If the distance between parallel walls is 'd' then required focal length of lens placed in between the walls is :–

(1) Only $\frac{d}{4}$

(2) Only $\frac{d}{2}$

(3) More than $\frac{d}{4}$ but less than $\frac{d}{2}$

(4) Less than or equal to $\frac{d}{4}$

PRI289

12. A short linear object of length L lies on the axis of a spherical mirror of focal length f at a distance u from the mirror. Its image has an axial length L' equal to

(1) $L\left[\frac{f}{(u-f)}\right]^{1/2}$ (2) $L\left[\frac{u+f}{f}\right]^{1/2}$

(3) $L\left[\frac{u-f}{f}\right]^{2}$ (4) $L\left[\frac{f}{(u-f)}\right]^{2}$

PRI290

13. An object is placed at a distance of 10 cm from a co-axial combination of two lenses A and B in contact. The combination forms a real image three times the size of the object. If lens B is concave with a focal length of 30 cm, the nature and focal length of lens A is

(1) convex, 12 cm

(2) concave, 12 cm

(3) convex, 6 cm

(4) convex, 18 cm

PRI291

Exercise-I ANSWER KEY

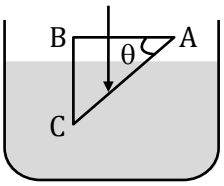
Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	2	2	4	4	2	2	3	2	1	3	4	4	3

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Exercise - II

Ray Optics and Optical Instruments

1. A glass prism ($\mu = 1.5$) is dipped in water ($\mu = 4/3$) as shown in figure. A light ray is incident normally on the surface AB. It reaches the surface BC after totally reflected, if



(1) $\sin\theta > 8/9$ (2) $2/3 < \sin\theta < 8/9$

(3) $\sin\theta \leq 2/3$ (4) It is not possible

PRI292

2. An object is seen through a simple microscope of focal length 12 cm. Find the angular magnification produced if the image is formed at the near point of the eye which is 25 cm away from it.

(1) $\frac{37}{6}$ (2) $\frac{37}{12}$ (3) $\frac{25}{12}$ (4) $\frac{25}{6}$

PRI293

3. In a compound microscope distance between both the lenses is 19 cm and focal length of objective and eyepiece is 2 cm and 3 cm respectively. Find out magnifying power of compound microscope when final image is formed at infinity.

(1) $-\frac{125}{3}$ (2) $-\frac{150}{3}$

(3) $-\frac{175}{3}$ (4) $-\frac{175}{6}$

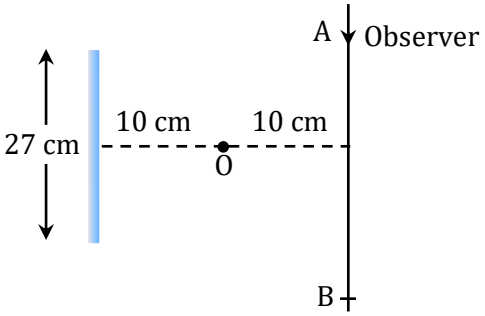
PRI294

4. Diameter of the moon is 3.5×10^3 km and its distance from earth is 3.8×10^5 km. It is seen by a telescope whose objective and eyepiece have focal lengths 4 m and 10 cm respectively. What will the angular diameter of the image of the moon in normal adjustment?

(1) 21° (2) 36° (3) 7° (4) 28°

PRI295

5. An observer is moving along the line AB as shown. When image of object O is first visible to observer, he starts from rest with the acceleration of 2cm/sec^2, then the time for which image is visible to observer is :-



(1) 3 sec (2) 9 sec

(3) 12 sec (4) None of these

PRI296

6. A plane mirror is moving with velocity $4\hat{i}+5\hat{j}+8\hat{k}$. A point object in front of the mirror moves with a velocity $3\hat{i}+4\hat{j}+5\hat{k}$. Here $\hat{i}$ is along the normal to the plane mirror and facing towards the object. The velocity of the image is :-

(1) $5\hat{i}+4\hat{j}+5\hat{k}$ (2) $5\hat{i}+4\hat{j}+8\hat{k}$

(3) $3\hat{i}+4\hat{j}+5\hat{k}$ (4) $6\hat{i}+4\hat{j}+5\hat{k}$

PRI297

7. An object is placed at 10 cm, in front of a concave mirror of radius of curvature 15 cm. Find the position, nature, and magnification of the image in each case.

(1) 30 cm, real, enlarged and inverted

(2) 30 cm, real, diminished and inverted

(3) 30 cm, virtual, diminished and erect

(4) 30 cm, virtual, enlarged and erect

PRI298

8. A point object is kept between a plane mirror and a concave mirror facing each other. The distance between the mirrors is 22.5 cm. Plane mirror is placed perpendicular to principal axis of concave mirror. The radius of curvature of the concave mirror is 20 cm. If distance of the object from the concave

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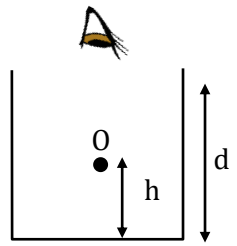
mirror is 15 cm. Then find out the position of image formed after two successive reflection (Consider first reflection from concave mirror)
(1) 30 cm from concave mirror
(2) 15 cm from plane mirror
(3) On the object itself
(4) None of these

PRI299

9. An opaque cylindrical tank with an open top has a diameter of 3.00 m and is completely filled with water. When the setting sun reaches an angle of 37° above the horizon, sunlight ceases to illuminate any part of the bottom of the tank. How deep is tank in meter?
(1) 4m (2) 5m (3) 6m (4) 3m

PRI300

10. A vessel having perfectly reflecting plane bottom is filled with water (μ = 4/3) to a depth d. A point source of light is placed at a height h above the reflecting surface (inside the water). Find the distance of final image from water surface as seen by the observer.



(1) $\frac{3}{4}(d+h)$ (2) $\frac{3}{4}(d-h)$
(3) Both (1) and (2) (4) None of these

PRI301

11. A ray of light is incident on a parallel slab of thickness t and refractive index n. If the angle of incidence i is small, then the lateral displacement between incident and emergent ray will be:
(1) $it\left(1-\frac{1}{n}\right)$ (2) $\frac{ti}{n}$
(3) $\frac{i}{t}\left(1-\frac{1}{n}\right)$ (4) None

PRI302

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	1	2	3	1	2	1	1	3	1	3	1

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Exercise - I Wave Optics

1. Two coherent sources of equal intensities produce a maximum of 100 units. If the amplitude of one of the sources is reduced by 20%, then the maximum intensity produced will be : -
(1) 100 (2) 81 (3) 89 (4) 60
PWO219

2. In an interference pattern of two waves fringe width is β. If the frequency of source is doubled then fringe width will become :
(1) $\frac{1}{2}\beta$ (2) β (3) 2β (4) $\frac{3}{2}\beta$
PWO220

3. In an interference pattern the $(n+4)^{th}$ blue bright fringe and n^{th} red bright fringe are formed at the same spot. If red and blue light have the wavelength of 7800 Å and 5200 Å then value of n should be :
(1) 2 (2) 4 (3) 6 (4) 8
PWO221

4. In YDSE d = 1mm, λ = 500 nm D = 1.0 m. How many Bright fringes are formed in one cm of screen.
(1) 20 (2) 25 (3) 10 (4) 15
PWO222

5. Direction of the first secondary maximum in the Fraunhofer diffraction pattern at a single slit is given by (a is the width of the slit) :
(1) $a\sin\theta = \frac{\lambda}{2}$ (2) $a\cos\theta = \frac{3\lambda}{2}$
(3) $a\sin\theta = \lambda$ (4) $a\sin\theta = \frac{3\lambda}{2}$
PWO223

6. A light has amplitude A and angle between analyser and polariser is 60°. Light is transmitted by analyser has amplitude.
(1) $A\sqrt{2}$ (2) $\frac{A}{\sqrt{2}}$ (3) $\frac{\sqrt{3}A}{2}$ (4) $\frac{A}{2\sqrt{2}}$
PWO224

7. Two Nicols are oriented with their principal planes making an angle of 60°. The percentage of incident unpolarized light which passes through the system is :
(1) 50% (2) 100%
(3) 12.5% (4) 37.5%
PWO225

8. Two waves $Y_1 = a\sin\omega t$ and $Y_2 = a\sin(\omega t+\delta)$ are producing interference, then resultant intensity is proportional to –
(1) $a\cos^2\delta/2$ (2) $a^2\cos\delta/2$
(3) $a^2\cos\delta$ (4) $a^2\cos^2\delta/2$
PWO226

9. In the Young's double slit experiment the central maxima is observed to be I_0. If one of the slits is covered, then intensity at the central maxima will become :
(1) $I_0/2$ (2) $I_0/\sqrt{2}$ (3) $I_0/4$ (4) I_0
PWO227

10. Calculate angular width in a single slit diffraction of central maxima if λ = 6000 Å, a = 18×10^{-5} cm : $\left[\sin^{-1}\left(\frac{1}{3}\right) = 19.47°\right]$
(1) 28.4° (2) 38.94°
(3) 30° (4) 260°
PWO228

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	2	1	4	1	4	4	3	4	3	2

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Exercise - II — Wave Optics

1. The intensity of the central fringe obtained in the interference pattern due to two identical slit sources is I. When one of the slits is closed then the intensity at the same point is I_0. Then the correct relation between I and I_0 is :
(1) $I = I_0$ (2) $I = 2I_0$
(3) $I = 4I_0$ (4) $I = I_0/4$
PWO229

2. In double slit experiment, the angular width of the fringes is 0.20° for the sodium light (λ = 5890Å). In order to increase the angular width of the fringes by 10%, the necessary change in the wavelength is :
(1) Increase of 589Å
(2) Decrease of 589Å
(3) Increase of 6479Å
(4) Zero
PWO230

3. In YDSE, λ = 500 nm, d = 1.00 mm and D = 1.0 m. Find minimum distance from the central maxima for which the intensity is half of the maximum intensity.
(1) 2.5×10^{-4} m (2) 1.25×10^{-4} m
(3) 3×10^{-6} m (4) 4.3×10^{-5} m
PWO231

4. If diffraction occurs through a single slit then intensity of first secondary maxima become ..... % of central maxima :-
(1) 4% (2) 25%
(3) 75% (4) 50%
PWO232

5. Two waves of intensity ratio 1 : 9 cross each other at a point. The resultant intensities at the point, when (a) Waves are incoherent is I_1(b) Waves are coherent is I_2 and differ in phase by 60°. If $\frac{I_1}{I_2} = \frac{10}{x}$ then x = ____.
(1) 11 (2) 21 (3) 13 (4) 41
PWO233

6. When unpolarized light is incident at an angle of 60° on a transparent medium from air. The reflected ray is completely polarized. The angle of refraction in the medium is
(1) 30° (2) 60°
(3) 90° (4) 45°
PWO234

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6
Answer	3	1	2	1	3	1

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Exercise - I Modern Physics-I

1. A laser beam (λ = 633 nm) has an power of 3 mW. What will be the pressure exerted on a surface by this beam if the cross-sectional area is 3 mm^2. (Assume perfect reflection and normal incidence)
(1) 6.6×10^{-3} N/m^2
(2) 6.6×10^{-6} N/m^2
(3) 6.6×10^{-9} N/m^2
(4) 6.6 N/m^2

PMD289

2. The frequency of the incident light falling on a photosensitive metal plate is doubled, the kinetic energy of the emitted photoelectrons is
(1) Double of the earlier value
(2) Unchanged
(3) More than double
(4) Less than double

PMD290

3. The threshold wavelength of tungsten is 2300 Å. If ultra violet light of wavelength 1800 Å is incident on it, then the maximum kinetic energy of photoelectrons would be.
(1) 1.5 eV (2) 2.2 eV
(3) 3.0 eV (4) 5.0 eV

PMD291

4. The kinetic energy of most energetic electrons emitted from a metallic surface is doubled when the wavelength of the incident radiation is changed from 400 nm to 310 nm the work-function of the metal is :
(1) 0.9 eV (2) 1.7 eV
(3) 2.2 eV (4) 3.1 eV

PMD292

5. If the light of wavelength λ is incident on metal surface, the ejected fastest electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be
(1) Smaller than $\sqrt{\frac{4}{3}}$ v
(2) Greater than $\sqrt{\frac{4}{3}}$ v
(3) 2 v
(4) Zero

PMD293

6. Threshold wavelength for photoelectric emission from a metal surface is 5200 Å. Photoelectrons will be emitted when this surface is illuminated with monochromatic radiation from.
(1) 1 W IR lamp (2) 50 W UV lamp
(3) 50 W IR lamp (4) 10 W IR lamp

PMD294

7. The work functions for metals A, B and C are respectively 1.92 eV, 2.0 eV and 5eV. According to Einstein's equation, the metals which will emit photo electrons for a radiation of wavelength 4100Å is/are : -
(1) None
(2) A only
(3) A and B only
(4) All the three metals

PMD295

8. Light of frequency 10^{15} Hz falls on a metal surface of work function 2.5 eV. The stopping potential of photoelectrons in volts is :-
(1) 1.6 (2) 2.5 (3) 4.1 (4) 6.6

PMD296

9. Work function of potassium metal is 2.30 eV. When light of frequency 8×10^{14} Hz is incident on the metal surface, photoemission of electrons occurs. The stopping potential of the electrons will be equal to
(1) 0.1 V (2) 1.0 V
(3) 2.3 V (4) 3.3 V

PMD297

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10. A proton moves on a circular path of radius 6.6×10^{-3} m in a perpendicular magnetic field of 0.625 tesla. The De broglie wavelength associated with the proton will be :

(1) 1Å (2) 0.1Å

(3) 0.01Å (4) 0.001Å

PMD298

11. The velocity at which the mass of a particle becomes twice of its rest mass, will be -

(1) $\frac{2c}{3}$ (2) $\frac{c}{2}$ (3) $\frac{c\sqrt{3}}{2}$ (4) $\frac{3c}{4}$

PMD299

12. An electron with (rest mass m_0) moves with a speed of 0.8C. Its mass when it moves with this speed is :

(1) m_0

(2) $\frac{m_0}{6}$

(3) $\frac{5m_0}{3}$

(4) $\frac{3m_0}{5}$

PMD300

Exercise-I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	2	3	1	3	2	2	3	1	2	3	3	3

AIM: FREE EDUCATION TO ALL APUL

Exercise - II Modern Physics-I

1. When a certain metallic surface is illuminated with mono-chromatic light of wave length λ, the stopping potential for photo electric current is 6 V_0. When the same surface is illuminated with light of wave length 2λ, the stopping potential is $2V_0$. The threshold wavelength of this surface for photoelectric effect is :

(1) 6λ (2) $4\lambda/3$ (3) 4λ (4) 8λ

PMD301

2. Two separate monochromatic light beams A and B of the same intensity are falling normally on a unit area of a metallic surface. Their wavelength are λ_A and λ_B respectively. Assuming that all the incident light is used in ejecting the photoelectrons, the ratio of the number of photoelectrons from the beam A to that from B is

(1) $(\lambda_A/\lambda_B)^2$ (2) λ_A/λ_B

(3) λ_B/λ_A (4) 1

PMD302

3. A photosensitive metallic surface has work function, $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface, the electrons come out with a maximum velocity of 4×10^6 m/s. When the photon energy is increased to $5h\nu_0$, then maximum velocity of photo electrons will be :-

(1) 2×10^7 m/s (2) 2×10^6 m/s

(3) 8×10^5 m/s (4) 8×10^6 m/s

PMD303

4. A 500 watt bulb is placed at the centre of a perfectly black sphere of radius R = 1 metre. The approximate pressure experienced by the walls of the sphere as it absorbs all the photon emitted by the bulb is (take $4\pi = 12.6$)

(1) 1.3×10^{-7} N/m^2

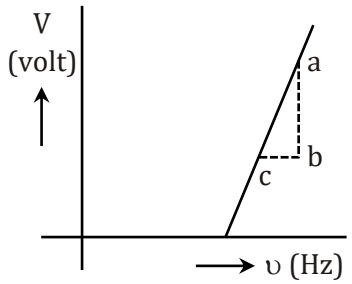
(2) 2.5×10^{-7} N/m^2

(3) 6.3×10^{-7} N/m^2

(4) 3.14×10^{-7} N/m^2

PMD304

5. In a photoelectric experiment the graph of frequency υ of incident light (in Hz) and stopping potential V (in volt) is shown below. From figure the value of the Planck's constant is (e is the elementary charge)



(1) $e\frac{ab}{cb}$ (2) $e\frac{cb}{ab}$

(3) $e\frac{ac}{bc}$ (4) $e\frac{ac}{ab}$

PMD305

Exercise-II ANSWER KEY

Question	1	2	3	4	5
Answer	3	2	4	1	1

AIM: FREE EDUCATION TO ALL APUL

Exercise - I — Modern Physics-II

1. The amount of U^{235} to be fissioned, to operate 10 kW nuclear reactor is (Approximately)
(1) 1.2×10^{-5} g/s (2) 1.2×10^{-7} g/s
(3) 1.2×10^{-9} g/s (4) 1.2×10^{-13} g/s
PMD306

2. A source of energy of 100 W is producing energy by fission of 1 kg U^{235}. How long it can kept generation of energy :- (approx)
(1) 2.5×10^{4} yr (2) 10^{6} s
(3) 8.6×10^{7} s (4) 100 yr
PMD307

3. Energy released by 1 kg. U^{235}, when it is fissoned.
(1) 8×10^{10} kWH (2) 5×10^{30} eV
(3) 10^{10} Joule (4) 5×10^{26} MeV
PMD308

4. Binding energy of deuterium is 2.23 MeV, then its mass defect in a.m.u. is :-
(1) –0.0024 (2) –0.0012
(3) 0.0012 (4) 0.0024
PMD309

5. 200 MeV of energy can be obtained by per fission. In a reactor generating 1000 kW find the number of nuclei under going the fission per second :-
(1) 1000 (2) 2×10^{8}
(3) 3.125×10^{16} (4) 931
PMD310

6. The total energy of an electron is 3.555 MeV, then its Kinetic energy is :-
(1) 3.545 MeV (2) 3.045 MeV
(3) 3.5 MeV (4) None
PMD311

7. Radius of nucleus varies as $R = R_0(A)^{1/3}$, where R_0 = 1.3 fermi. What is the volume of Be^{8} nucleus (approx) [A = atomic mass]
(1) 7×10^{-38} cc (2) 7×10^{-29} cc
(3) 7×10^{-45} cc (4) none of the above
PMD312

8. Fission of one nucleus $_{92}U^{235}$ releases 250 MeV of energy. The number of fissions per second required to produce 1 MW power is
(1) 9.2×10^{17} (2) 6.3×10^{23}
(3) 1.6×10^{19} (4) 2.5×10^{16}
PMD313

9. The radius of a spherical nucleus as measured by electron scattering is 3.6 fm. What is the likely mass number of the nucleus?
(1) 27 (2) 40
(3) 56 (4) 120
PMD314

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	2	1	4	4	3	2	1	4	1

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Exercise - II Modern Physics-II

1. A nuclear fusion reaction is given below :
$_1H^2 + _1H^2 \rightarrow _2He^3 + n + 3.2$ MeV
How much energy will be generated when 2 kg of deuterons are fused :- (approx)
(1) 10^{30} eV (2) 5×10^{23} MeV
(3) 10^{22} MeV (4) 10^{33} eV
PMD315

2. A γ ray photon produces an electron positron pair, If the rest mass energy of electron is 0.51 Mev and the total kinetic energy of electron-positron pair is 0.78 MeV then the energy of γ -ray photon in Mev is :-
(1) 0.78 (2) 1.8
(3) 1.28 (4) 0.28
PMD316

3. We are given the following atomic masses
^{238}U = 238.05079 u ^{234}Th = 234.04363 u
^4He = 4.00260 u
The energy released during the alpha decay of ^{238}U is :-
(1) 6.00 MeV (2) 4.25 MeV
(3) 3.75 MeV (4) 5.03 MeV
PMD317

4. A $_6C^{12}$ nucleus is to be divided into 3 alpha particles. The amount of energy required to achieve this (mass of an alpha particle = 4.00388 u) is (Mass of $_6C^{12}$ atom = 12.0000 u)
(1) 3. 405 MeV (2) 10.837 MeV
(3) 8.133 MeV (4) 12.573 MeV
PMD318

5. The binding energy of deuteron $\left(_1^2H\right)$ is 1.15 MeV per nucleon and an alpha particle has binding energy of 7.1 MeV per nucleon. Then in the reaction $_1^2H + _1^2H \rightarrow _2^4He + Q$ the energy Q is
(1) 33. 0 MeV (2) 28.4 MeV
(3) 23.8 MeV (4) 4.6 MeV
PMD319

6. Consider the following reaction
$_1^1H + _1^3H \rightarrow _1^2H + _1^2H$
The atomic masses are given as
$m\left(_1^1H\right)$=1.007825 u $m\left(_1^2H\right)$= 2.014102 u
$m\left(_1^3H\right)$ = 3.016049 u
The Q– value of the above reaction will be
(1) –4.03 MeV (2) –2.01 MeV
(3) 2.01 MeV (4) 4.03 MeV
PMD320

7. What is energy released in the β-decay of $^{32}P \rightarrow ^{32}S$? (Given : atomic masses : 31.97391 u for ^{32}P and 31.97207 u for ^{32}S)
(1) – 1.2 MeV (2) + 1.7 MeV
(3) + 2.1 MeV (4) – 0.9 MeV
PMD321

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	4	2	2	2	3	1	2

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Exercise - I — Modern Physics-III

1. The ionization potential of the hydrogen atom is 13.6 V. The energy needed to ionize a hydrogen atom which is in its first excited state is about
(1) 13.6 eV
(2) 10.2 eV
(3) 3.4 eV
(4) 1.5 eV

PMD322

2. If electron in a hydrogen atom has moved from n = 1 to n = 10 orbit, the potential energy of the system has
(1) increased
(2) decreased
(3) remained unchanged
(4) become zero

PMD323

3. The radius of third stationary orbit of electron for Bohr's atom is R. The radius of fourth stationary orbit will be:

(1) $\frac{4}{3}R$ (2) $\frac{16}{9}R$

(3) $\frac{3}{4}R$ (4) $\frac{9}{16}R$

PMD324

4. If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where α = _____.

(1) 144 (2) 145
(3) 245 (4) 241

PMD325

Exercise-I — ANSWER KEY

Question	1	2	3	4
Answer	4	1	2	1

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Exercise - II Modern Physics-III

1. Energy levels A, B and C of a certain atom correspond to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are wave lengths of radiations corresponding to transitions C to B, B to A and C to A respectively, which of the following relations is correct: –

(1) $\lambda_3 = \lambda_1 + \lambda_2$

(2) $\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$

(3) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(4) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

PMD326

2. Ionization potential of hydrogen is 13.6 volt. If it is excited by a photon of energy 12.1 eV, then the number of lines in the emission spectrum will be

(1) 2 (2) 3 (3) 4 (4) 5

PMD327

3. The electron of a hydrogen atom revolves round the proton in a circular n^{th} orbit of radius $r_n = \frac{\varepsilon_0 n^2 h^2}{(\pi m e^2)}$ with a speed $\upsilon_n = \frac{e^2}{2\varepsilon_0 nh}$.

The current due to the circulating charge is proportional to

(1) e^2 (2) e^3 (3) e^5 (4) e^6

PMD328

Exercise-II ANSWER KEY

Question	1	2	3
Answer	2	2	3

Exercise - I

Semiconductor Electronics

1. What will be conductivity of pure silicon crystal at 300 K temperature. If electron hole pairs per cm^3 is 1.072×10^{10} at this temperature, $\mu_e = 1350\ cm^2/volt\text{-}s$ and $\mu_h = 480\ cm^2/volt\text{-}s$:

(1) 3.14×10^{-6} mho/cm

(2) 3×10^{6} mho/cm

(3) 10^{-6} mho/cm

(4) 10^{6} mho/cm

PSE226

2. In sample of pure silicon 10^{13} atom/cm^3 is mixed of phosphorus. If all donor atoms are active then what will be resistivity at 20°C if mobility of electron is 1200 cm^2/volt-s :–

(1) 0.5209 ohm cm (2) 5.209 ohm cm

(3) 52.09 ohm cm (4) 520.9 ohm cm

PSE227

3. Mobility of electrons in N-type Ge is 5000 cm^2/volts and conductivity 5 mho/cm. If effect of holes is negligible then impurity concentration will be : –

(1) $6.25 \times 10^{15}/cm^3$ (2) $9.25 \times 10^{14}/cm^3$

(3) $6 \times 10^{13}/cm^3$ (4) $9 \times 10^{13}/cm^3$

PSE228

4. The energy of radiation emitted by LED is :-

(1) greater than the band gap of the semiconductor used.

(2) always less than the band gap of the semiconductor used.

(3) always equal to the band gap of the semiconductor used.

(4) equal to or less than the band gap of the semiconductor used.

PSE229

5. Two similar blocks of intrinsic silicon are separately doped uniformly to obtain same dopant densities $N_D = N_A$ per m^3. The n-type and p-type samples thus obtained are arranged in parallel to each other and a current is passed through the combination Then

(1) same current passes through each sample

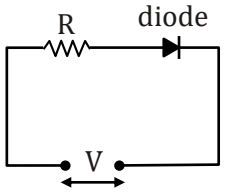
(2) current through n-type sample is more

(3) current through p-type sample is more

(4) current through p-type sample is twice that through the n-type sample

PSE230

6. For the given circuit of p-n junction diode which is correct :–



(1) in F.B. the voltage across R is V

(2) in R.B. the voltage across R is V

(3) in F.B. the voltage across R is 2 V

(4) in R.B. the voltage across R is 2 V

PSE231

7. The intrinsic carrier density in germanium crystal at 300 K is 2.5×10^{13} per cm^2. If electron density in an n-type germanium crystal at 300 K be 0.5×10^{17} per cm^3, a hole density (per cm^3) in this n-type crystal at 300 K would be expected around

(1) 2.5×10^{13} (2) 5×10^{6}

(3) 1.25×10^{10} (4) 0.2×10^{4}

PSE232

8. In half wave rectifier peak value of sinusodial signal is 10 V. Determine D.C. component at output :–

(1) $\frac{10}{\sqrt{2}}$V (2) $\frac{10}{\pi}$V

(3) 10 V (4) $\frac{20}{\pi}$V

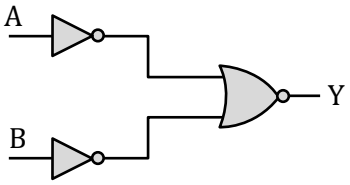
PSE233

9. The number of charge carrier concentration in silicon at 300 K is $1.56 \times 10^{16}/m^3$. If the electron and hole mobilities are respectively 0.135 and 0.065 (SI units) the resistivity of the material is (in SI units) approximately: -

(1) 2500 (2) 2000 (3) 3600 (4) 1350

PSE234

10. Which logic gate is represented by the following combination of logic gate: -



(1) OR (2) NAND
(3) AND (4) NOR

PSE235

11. Given truth table is related with :–

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

(1) NOT Gate (2) OR Gate
(3) XOR Gate (4) NAND Gate

PSE236

12. Two NOT gates are connected at the two inputs of a NAND gate. This combination will behave like **:-**

(1) NAND gate
(2) AND gate
(3) OR gate
(4) NOR gate

PSE237

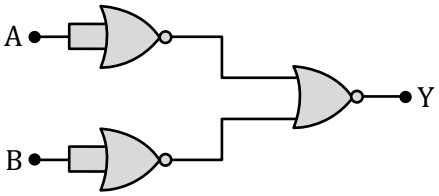
13. The truth table given below belongs for which gates?

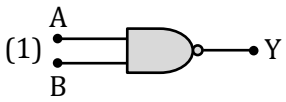
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

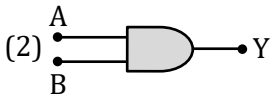
(1) OR gate
(2) XOR gate
(3) AND gate
(4) NAND gate

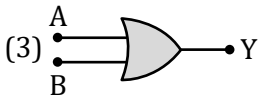
PSE238

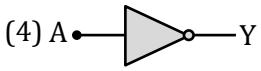
14. The following combination of gates is equivalent to :-



(1) 

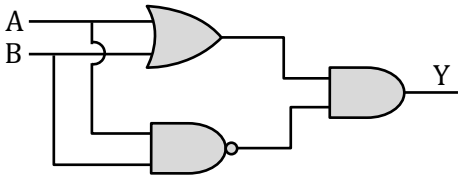
(2) 

(3) 

(4) 

PSE239

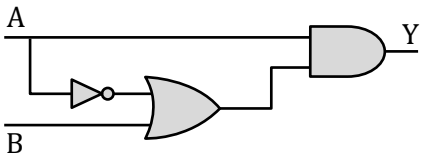
15. The following configuration of gates is equivalent to:-



(1) NAND (2) OR
(3) XOR (4) NOR

PSE240

16. The combination of the gates shown represents: –



(1) AND gate (2) OR gate
(3) NAND gate (4) NOR gate

PSE241

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	4	2	1	3	2	2	3	4	3	2	2	3
Question	16														
Answer	1														

Exercise - II

Semiconductor Electronics

1. An electric field is applied to a semiconductor. Let the number of charge carriers density is 'n' and the average drift speed be v. If the temperature is increased : –
(1) both n and v will increase
(2) n will increase but v will decrease
(3) v will increase but n will decrease
(4) both n and v will decrease

PSE242

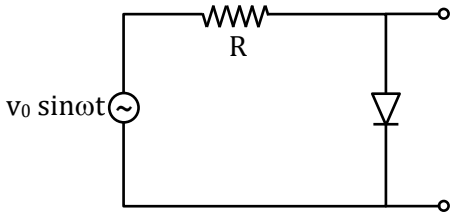
2. If the two ends of a p-n junction are joined by a wire : –
(1) there will not be a steady current in the circuit.
(2) there will be a steady current from the n-side to the p-side.
(3) there will a steady current from the p-side to the n-side.
(4) there may or may not be a current depending upon the resistance of the connecting wire.

PSE243

3. A hole diffuses from the p-side to the n-side in a p-n junction. This means that : –
(1) a bond is broken on the n-side and the electron freed from the bond jumps to the conduction band.
(2) a conduction electron on the p-side jumps to a broken bond to complete it.
(3) a bond is broken on the n-side and the electron free from the bond jumps to a broken bond on the p-side to complete it.
(4) a bond is broken on the p-side and the electron free from the bond jumps to a broken bond on the n-side to complete it.

PSE244

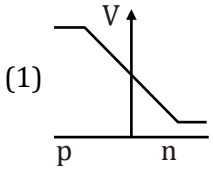
4. The output across the diode in the given circuit in figure.

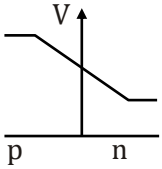


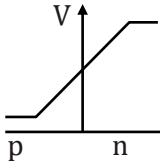
(1) Would be zero at all times.
(2) Would be like a half wave rectifier with positive cycles in output.
(3) Would be like a half wave rectifier with negative cycles in output.
(4) Would be like that of a full wave rectifier.

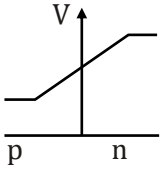
PSE245

5. In a forward biased p-n junction diode, the potential barrier in the depletion region is of the form : –

(1) 

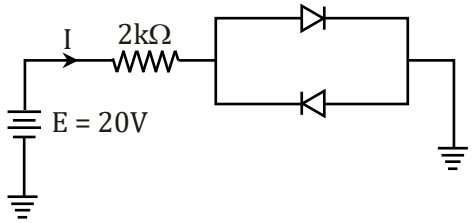
(2) 

(3) 

(4) 

PSE246

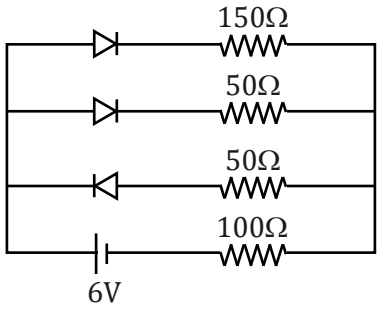
6. Assuming the diodes to be of silicon with forward resistance zero, the current I in the following circuit is :-



(1) 0
(2) 9.65 mA
(3) 10 mA
(4) 10.36 mA

PSE247

7. The circuit shown in the figure contains three diodes each with forward resistance of 50 ohms and with infinite backward resistance. If the battery voltage is 6 V, the current through the 100 ohm resistance is :-

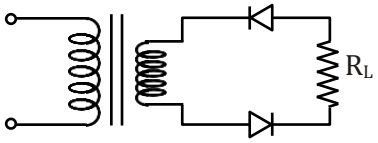


(1) 0
(2) 36 mA
(3) 43 mA
(4) 50 mA

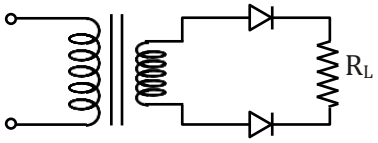
PSE248

8. Which of the following is not a rectifier circuit?

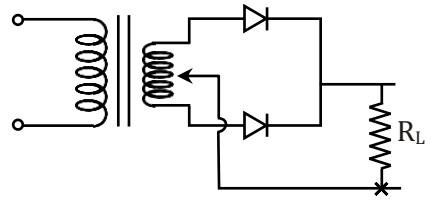
(1)



(2)



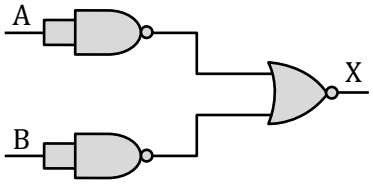
(3)



(4) None of these

PSE249

9. The truth table for the gate shown in the following figure will be :-



(1)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

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10. The output of gate is low when at least one of its input is high. This is true for: –

(1) NOR (2) OR

(3) AND (4) NAND

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11. 'Output is LOW if and only if all the inputs are HIGH'

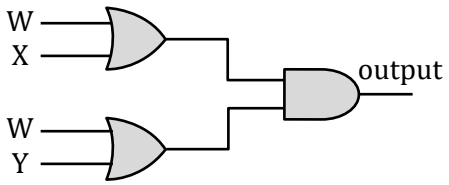
Indicate the logic gate for which the above statement in true :–

(1) AND (2) OR

(3) NOR (4) NAND

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12. The diagram of a logic circuit is given below. The output of the circuit is represented by :–



(1) W. (X + Y) (2) X. (X.Y)

(3) W + (X + Y) (4) W + (X.Y)

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Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	2	1	3	3	4	2	2	2	4	1	4	4

Important Notes